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Fully reconfigurable Mach Zehnder interferometers for the storage of photonic entanglement in a laser written photonic chip

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Abstract

Quantum memories are important tools in quantum photonics, in which they allow the syncrhronization of multiple devices when many different probabilistic quantum processes are linked together. The realization of efficient, long-lived quantum memories is of paramount importance for many applications, for example the creation of large scale secure quantum networks.
This thesis work aimes the realization of a pair of integrated photonic waveguide circuits containing reconfigurable Mach Zehnder interferometers compatible with solid state quantum memories for the construction of a hybrid glass-crystal integrated platform capable of storing photonic entanglement. The interferometers are fabricated by femtosecond laser micromachining in borosilicate glass. One interferometer will be placed before the quantum memories to create the path entanglement, while the other will be placed after the memories and it will allow to carry out tomography of the output state.
Each interferometer is equipped with two microheaters patterned on top of it, to allow for the control of the internal and external phases exploiting the thermo-optic effect. Isolation microstructures are realized by laser ablation beneath each microheater to improve the heating efficiency.
The devices have been optimized to work at a wavelength of 607 nm to be used together with the already available quantum memories. Propagation losses as low as 0.13 dB/cm are obtained for the optical waveguides, together with good mode matching with commercial single-mode fibers.
The final device can be used to store path-encoded photonic qubits following the Atomic Frequency Comb protocol for operating the quantum memory.

Keywords: quantum memory, femtosecond laser micromachining, integrated optical circuit, reconfigurability

Riassunto

Le memorie quantistiche sono un'importante stromento per la fotonica quantistica, poichè permettono di sincronizzare dispositivi diversi in cui avvengono processi probabilistici. La realizzazione di memorie quantistiche efficienti e con lunghi tempi di coerenza è di primaria importanza per molte applicazioni, tra cui la creazione di network quantistici sicuri su larga scala.
Questa tesi è mirata alla realizzazione di una coppia di circuiti fotonici integrati contentnti interferometri Mach Zehnder riconfigurabili, da usare assieme a una coppia di memorie quantistiche a stato solido per l'immagazzinamento entanglement fotonico. Gli interferometri sono fabbricati tramite scrittura laser a femtosecondi in chip di vetro borosilicato. Un interferometro verrà posizionato prima delle memorie quantistiche per la preparazione dello stato da immagazzinare, mentre l'altro seguirà le memorie per condurre una tomografia sullo stato in uscita da esse.
Ogni interferometro è dotato di due microriscaldatori per il controllo delle fasi interne ed esterne tramite effetto termo-ottico. Per rendere più efficiente la dissipazione di calore, sono state realizzate microstrutture di isolamento sotto ai microriscaldatori.
I dispositivi sono stati ottimizzati per lavorare a 607 nm, così da essere usati assieme alle memorie quantistiche, già disponibili. Le strutture guidanti mostrano perdite di propagazione fino a 0.13 dB/cm e buon *mode-matching* con fibre ottiche a singolo modo in commercio.
Il dispositivo globale può essere usato per l'immagazzinamento di qubit fotonici *path-encoded* secondo lo schema AFC - *atomic frequency comb*.

Parole chiave: memorie quantistiche, scrittura laser a femtosecondi, circuito ottico integrato, riconfigurabilità

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Introduction

Quantum technologies are affirming as a prominent perspective in information science. By exploiting the peculiar features of quantum mechanics, devices can be realized capable of controlling superposition and entanglement of individaual quantum particles to obtain performance advantages over classical machines. In this context, photons offer interesting features as carriers of quantum information, giving rise to a branch of quantum technologies known as quantum photonics. Photons have slow decoherence and are easy to manipulate with classical optical elemets such as for example mirrors, beam splitters and polarizers. Moreover, they do not interact with each other unless nonlinear effects come into play, which avoids unwanted crosstalks.

Quantum memories are key devices for quantum photonics. They allow to store a quantum state encrypted in a photon for later retrieval. Quantum memories can be employed in a variety of applications, such as linear-optical quantum computing, quantum-enhanced metrology and quantum communications. Solid state quantum memories rose to interest due to their compactness, scalability and possibility to be easily integrated with other optical circuits. While bulk devices offer - to date - better performances and are more easily accessible, integrated devices based on guided-wave optics offer great advantages in terms of stability, robustness. footprint and construction of a hybrid glass-crystal integrated platform capable of storing photonic entanglement.

Among other integrated-optics fabrication techniques, femtosecond laser micromachining has emerged as a versatile tool for the realization of photonic quantum circuits. This technology allows the inscription of high-quality waveguides in transparent materials in a fast and cost-effective way, making it a powerful technique for rapid prototiping of photonic quantum circuits since it does not require any lithographic step or clean room facility. Moreover, it is a true 3D technology, allowing to experiment on otherwise inaccessible geometries. Femtosecond laser-written circuits can also be easily interfaced to optical fibers, for the creation of low-loss, stable nodes for quantum networks.

This thesis work aims at the realization of two fully-reconfigurable Mach Zehnder interferometers by femtosecond laser micromachining. The interefeometers will be then glued

to a pair of waveguide quantum memories inscribed inside a rare-earth doped crystal, for the storage of path-encoded photonic qubits. This experiment has been conceived within the scientific collaboration between Fast group at IFN-CNR and the quantum photonics group led by prof. de Riedmatten at ICFO in Barcelona.

We will present the design and specification of the reconfigurable Mach Zehnder interferometers, describing the fabrication and characterization procedures in detail.
The thesis is structured as follows. In Chapter 1, we review the physical principles and main applications of quantum memories, briefly describing the state of the art of solid state devices. In Chapter 2, we describe the working principles of femtosecond laser micromachining, together with reviewing the present literature on its applications for quantum optics with special regard for reconfigurable devices. In Chapter 3, we describe the experimental setup and methods employed for the fabrication and characterization of the two interferometers. In Chapter 4, we discuss the design of the global intergferometer for the storage experiment and we present the characterization of both the passive interferometers and the thermo-optic phase shifters that allow their reconfigurability.
All the fabrication and characterization processes have been carried out at the department of physics of Politecnico di Milano.

1 | Integrated Quantum Memories

1.1. Introduction to quantum information

1.1.1. Qubit encoding

The quantum counterpart to the classical bit is the *qubit*. While a bit can only assume two values, 0 and 1, a qubit exists in a general superposition of two quantum states $|0\rangle$ and $|1\rangle$, which form the qubit computational basis. In general, we write

$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle \tag{1.1}$$

where the coefficients $\alpha,\ \beta \in \mathbf{C}$ are normalized, such that $|\alpha|^2 + |\beta|^2 = 1$.
Following Born's rule, $|\alpha|^2$ and $|\beta|^2$ represent the probabilities that a measurement of the qubit on the computational basis yields $|0\rangle$ or $|1\rangle$, respectively.

In fact, any two-level quantum system can encode a qubit. In particular, photons offer many degrees of freedom for qubit encoding. For example, one may use polarization encoding, temporal bin encoding or even path encoding, which is of interest for this work. Path encoding exploits the possibility of delocalizing a single photon in two different spatial modes *up* and *down*. Since it relies on the presence of two spatial modes, the path encoding scheme is also referred to as *dual rail* encoding, as opposed to *single rail* encoding schemes in which the two logic states occupy the same spatial mode - like polarization encoding.
In the context of this work, it's worth nothing that optical waveguides in integrated optics correspond to defined spatial modes.

1.1.2. Entanglement and Bell states

Quantum entanglement is one of the most counterintuitive features of quantum mechanics, reguarding the nonseparability of quantum states of systems that may not even be interacting [1–4].

Without loss of generality, we can restrict our discussion to two-particle - or *bipartite* - systems.
Mathemathically, the system is said to be *entangled* if its wavefunction can't be factorized as the product of its component states. For instance, consider two photons in polarization encoding, using horizontal $|H\rangle$ and vertical $|V\rangle$ polarization states as the computational basis. An example of two polarization-entangled photons is a system whose wavefunction can be written as

$$|\psi\rangle = \frac{1}{\sqrt{2}} \left(|H\rangle_A \otimes |V\rangle_B + |V\rangle_A \otimes |H\rangle_B\right) \tag{1.2}$$

where the subscripts A and B label the two photons. Physically, equation 1.2 means that if a measurement of the polarization of photon A yields horizontal polarization, then photon B will be found to have vertical polarization with certainty, and viceversa.
Notice how path encoding allows for the creation of entanglement over a single photon - indeed, the wavefunction of a photon delocalized on the two modes *up* and *down* can be written as

$$|\psi\rangle = \alpha\,|1\rangle_{up} \otimes |0\rangle_{down} + \beta\,|0\rangle_{up} \otimes |1\rangle_{down} \tag{1.3}$$

which means that a measurement will find either one photon on the upper path and none on the lower path, or none on the upper path and one on the upper path.
In general, the canonical computational base for entangled states of bipartite systems is the one composed by the four *Bell states*

$$\left|\Phi^{\pm}\right\rangle = \frac{1}{\sqrt{2}} \left(|0\rangle_A \otimes |0\rangle_B + |1\rangle_A \otimes |1\rangle_B\right) \tag{1.4}$$

$$\left|\Psi^{\pm}\right\rangle = \frac{1}{\sqrt{2}} \left(|0\rangle_A \otimes |1\rangle_B + |1\rangle_A \otimes |0\rangle_B\right) \tag{1.5}$$

where $|0\rangle$ and $|1\rangle$ represent the computational basis of the qubit, and A and B are the two particles.

1.1.3. Quantum teleportation

Quantum communication is a branch of quantum information that exploits the features of quantum mechanics to grant secure, hacker-proof links.
To do so, quantum states need to be transferred from one place to another.
Actually, copying a quantum state onto a system without destroying the original state is forbidden by the laws of quantum mechanics, a feature known as the *no-cloning theorem* [5]. A way around this is given by another peculiar phenomenon, known as *quantum teleportation*. Here, an unknown quantum state is transferred from one point to another,

those points possibly being widely separated in space. We emphasize on the fact that it's the quantum *state* that is teleported, not the particle(s) in such state.
Here we describe the phenomenon as originally given by Bennett et al. [6], denoting the general encoded states with $|0\rangle$ and $|1\rangle$.
The teleportation protocol involves two partecipants, usually called Alice (A) and Bob (B). Schematics of the protocol are shown in Figure 1.1. Suppose Alice is given a quantum

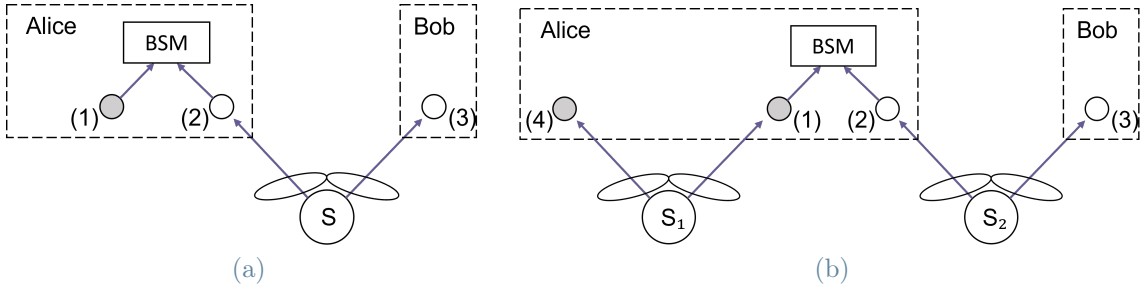


Figure 1.1: Quantum teleportation schemes. a) Quantum teleportation as described in [6]. Alice wishes to teleport the quantum state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ encoded in photon (1) to Bob. Alice and Bob share two entangled photons (2) and (3), produced by the source S. Alice performs a Bell state measurement (BSM) on photons (1) and (2), and then conveys the result of the BSM to Bob by classical means. Bob's photon (3) then collapses onto a defined state according to equation 1.8, which can be turned back to $|\psi\rangle$ by means of logical operations. b) Scheme of an entanglement swapping experiment. This time, photon (1) is in turn entangled with photon (4), which Alice possesses. After the teleportation protocol is realized, the state of (1) is transferred to (3), so that in the end the two photons (3) and (4) are entangled, even though they have no common origin - the entanglement is created by the BSM. Photons (4) and (3) may be stored inside quantum memories for later use - this configuration constitutes the basic node of a quantum repeater architecture (see Section 1.2.1).

state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, and she wants to teleport it to Bob so that in the end Bob possesses a photon in the same state $|\psi\rangle$.
Notice how the $|\psi\rangle$ is actually unknown to Alice, namely she doesn't know the coefficients α and β - had she known them, she could have easily conveyed them to Bob by classical means.
Suppose there exists a light source shared between Alice and Bob, able to produce two entangled photons whose state can be written as

$$|\Psi_{AB}\rangle = \frac{1}{\sqrt{2}}\left(|0\rangle_A \otimes |0\rangle_B + |1\rangle_A \otimes |1\rangle_B\right) \tag{1.6}$$

shared between Alice and Bob as indicated by the subscripts.
So far, Alice is in possession of the state to teleport $|\psi\rangle$ and her part of the shared entangled state, while Bob only has his part of the entangled state.
The total state for Alice and Bob can be written as

$$|\Phi_{AB}\rangle = |\psi\rangle\,|\Psi_{AB}\rangle = \frac{1}{\sqrt{2}}\left(c_0\,|0\rangle + c_1\,|1\rangle\right)\left(|0\rangle_A \otimes |0\rangle_B + |1\rangle_A \otimes |1\rangle_B\right) \tag{1.7}$$

which, remembering the Bell states introduced in 1.4, can be expanded as

$$\begin{aligned}|\Phi_{AB}\rangle &= \frac{1}{\sqrt{2}}(\alpha\,|0\rangle_A \otimes |0\rangle_B + \alpha\,|0\rangle\,|1\rangle_A \otimes |1\rangle_B + \beta\,|1\rangle\,|0\rangle_A \otimes |0\rangle_B + \beta\,|1\rangle\,|1\rangle_A \otimes |1\rangle_B)\\ &= \frac{1}{2}\Big[\left|\Phi^+\right\rangle(\alpha\,|0\rangle_B + \beta\,|1\rangle_B) + |\Phi^-\rangle(\alpha\,|0\rangle_B - \beta\,|1\rangle_B)\\ &\quad + \left|\Psi^+\right\rangle(\alpha\,|1\rangle_B + \beta\,|0\rangle_B) + |\Psi^-\rangle(\alpha\,|1\rangle_B - \beta\,|0\rangle_B)\Big]\end{aligned} \tag{1.8}$$

This way, we highlight how each Bell state on Alice's side is correlated to a certain superposition of the state in Bob's part of the shared state. Notice however how nothing physical has yet happened, all we have done was re-arrange equation 1.7
At this point, Alice performs a projective measurement on the Bell basis - a *Bell state measurement* (BSM) [7]. The outcome of the BSM is random, and each of the four Bell states has a 1/4 probability of occurring.
Suppose for instance Alice measures the Bell state $|\Psi^+\rangle$, and she knows she obtained that state. Then, according to equation 1.8 Bob's photon will be projected onto the state $\alpha\,|1\rangle_B + \beta\,|0\rangle_B$.
Now, Alice tells Bob over a classical channel that she obtained $|\Psi^+\rangle$, so that both know what state Bob's photon collapsed onto. To obtain the original state $|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle$, Bob only needs to perform a "flip" operation $|1\rangle_B \rightarrow |0\rangle_B$, $|0\rangle_B \rightarrow |1\rangle_B$, equivalent to a NOT logical gate.
A few remarks are necessary:

- If Alice got a different Bell state, Bob would only need perform a different logical operation to obtain the original state $|\psi\rangle$.
- At no time did anyone know the original state being teleported - the coefficients α and β are still unknown at the end of the protocol.
- Since the original photon is destroyed with the BSM, the no-cloning theorem is respected.
- In principle, any pure state can be teleported. For instance, Alice may teleport an

entangled state to Bob, a procedure known as *entanglement swapping* (see Figure 1.1b).

1.2. Quantum memories

Quantum memories (QMs) for photons, where non-classical states of light are mapped onto stationary matter states and preserved for subsequent retrieval, are important tools for quantum information science enabled by tight control over interactions between light and matter [8]. QMs are important devices in quantum information science because they can be used as synchronization devices when many different probabilistic quantum processes are linked together, by providing an interface between flying and stationary qubits.

Different technologies can be used to implement quantum memories, such as cold atomic ensambles [9], atomic vapors [10], single quantum systems [11–13], or criogenically-cooled rare-earth doped solids [14].

1.2.1. Main applications

QMs are required for the implementation of scalable protocols in quantum photonics, with potential applications in optical quantum computing, generation of multiphoton states from probabilistic pair sources [15], quantum information networks [16] and long-distance quantum communication using quantum repeaters [9, 17, 18].

Deterministic single photon sources

Given a non-deterministic source of photon pairs - e.g. based on spontaneous parametric down-conversion (SPDC) - a deterministic source of single photons can be realized by detecting one photon of a pair, while storing the other in a QM. The detection event of the first photon means the QM is charged, and can be used as a deterministic single-photon emitter. This requires the memory to be highly efficient. Deterministic single photon sources are necessary for linear optics quantum computing [19, 20], and are useful for quantum repeater protocols [18, 21].

Quantum repeaters and long-distance communications

QMs are necessary in the vast majority of the schemes devised for the realization of large-scale quantum communication networks, which make use of *quantum repeaters* [17, 21]. In principle, direct distribution of quantum states is limited by transmission losses and

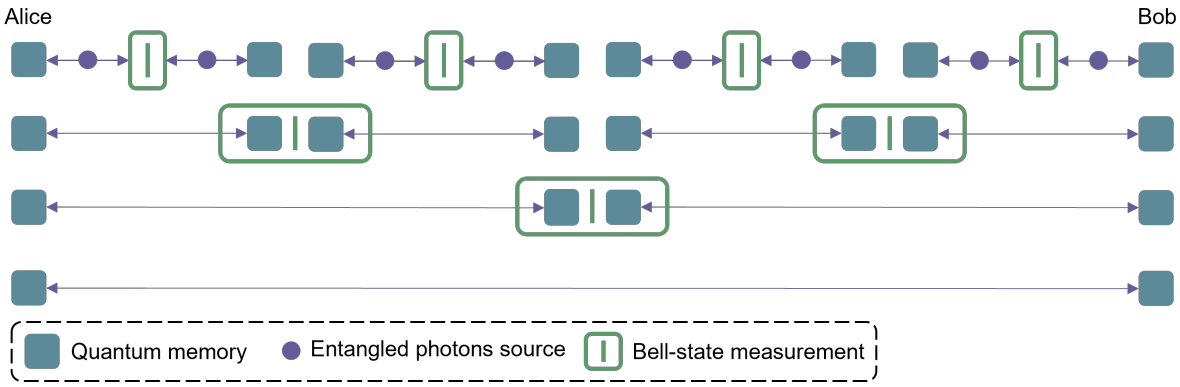


Figure 1.2: Operational principle of the quantum repeater architecture. Instead of performing entanglement swapping over a large distance, the communication is factorized into smaller links in which entanglement is created independently. Each link stores the entanglement by using quantum memories. Then, entanglement is propagated to the more external node until the target links are entangled.

the no-cloning theorem.
Following what we said in Section 1.1.3, a way around the impasse is to use quantum teleportation protocols. Quantum repeaters combine quantum teleportation with quantum state storage. The idea is to divide a long distance into shorter links, create and store entanglement independently in each link, then extend it to the whole distance via entanglement swapping.
Due to the probabilistic nature of entanglement swapping, this approach is only possible if QMs are available to synchronize all the different BSM steps.
The working principle of quantum repeaters is shown in Figure 1.2.

Loophole-free Bell tests

A loophole-free test of Bell's inequality [22] has two major requirements - entanglement must be created over a long distance to enforce locality, and high-efficiency detection of the entangled state is necessary to close the efficiency loophole.
Both this requirements can be solved by using highly efficient QMs in a similar scheme to that of a quantum repeater, to create entanglement between distant atoms via the detection of photons [23].

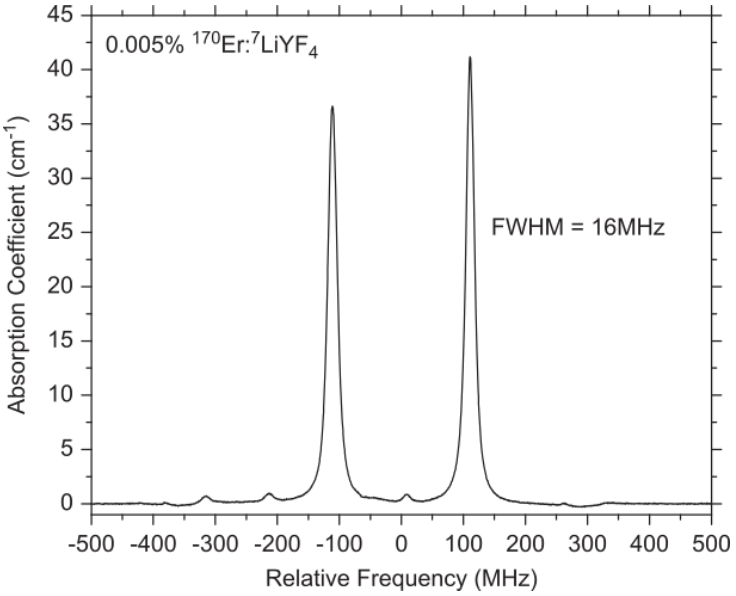


Figure 1.3: Laser absorption spectrum of 0.005% $^{170}Er^{3+}$:$^{7}LiYF_4$ showing ultra-narrow inhomogeneous absorption lines with FWHM= 16 MHz. Image adapted from reference [30].

Optical non-linearity mediated by storage

Creating nonlinear interactions at the single-photon level is a major challenge in quantum optics, requiring precise control of light-matter interfaces.
Quantum storage techniques come useful for mediating the photon-photon interaction, using different storage devices such as cold atoms [24], waveguides in solids [25], single trapped atoms [26] or Rydberg atoms [27–29].

In the following, we will focus on solid state devices, which are the final scope of this work.

1.2.2. Solid state quantum memories

Rare-earth ions (REI) doped solids are appealing candidates for implementing quantum information storage devices [30].
Due to the shielding of the 4f electron shell from outermost 5s and 5p electrons, the coupling of 4f electrons to the surrounding matrix lattice is strongly reduced. As a result, REIs in crystals are characterized by sharp homogeneous transition lines. In addition, such devices provide optical access to electronic and nuclear spin states, and demonstrate significant optical inhomogeneous broadening that can be tailored by optical pumping techniques.

The optical homogeneous linewidth can be extremely narrow - ranging from <1 kHz to 1 MHz at cryogenic temperatures - while the optical inhomogeneous linewidth varies strongly between crystal hosts, ranging from a few hundreds of MHz to tens of GHz. Since the homogeneous linewidth is ultimately limited by magnetic interactions - with other rare-earth ions or with magnetic constituents of the lattice - better performances are obtained for crystal materials with low nuclear spin concentration and low REI dopant concentration.
The REI's magnetic dipole moment also plays a fundamental role in the inhomogeneous broadening. In particular, REIs with an even number of electrons in the optical shell - also called *non-Kramers* ions - are preferred with respect to *Kramers* ions, which present an odd number of optical electrons leading to a large magnetic dipole. This narrows down the choice of the REI dopant to a few atomic species - for example, Pr^{3+}, Er^{3+} or Eu^{3+}.
Finally, being based on a solid-state platform, these system are naturally suitable to optical integration [31] to the micro- nano-scale [8].

Quantum memory protocols

Most quantum memory protocols are inspired by the photon echo process [32].
There, the optical pulse to be stored is absorbed by a inhomogeneous ensemble of atoms. The induced atomic coherences then undergo inhomogeneous dephasing, which can be reversed by applying an optical π-pulse after a time τ after the input pulse. After a time 2τ, the atomic coherences are back in phase, which results in the strong emission known as photon echo.
However, this simple protocol cannot be used for long-lived storing, since the recall π-pulse causes too much spontaneous emission noise due to the high atomic excitation density it induces [33].
A solution was proposed around 2005, when a scheme in which the dephasing is controlled by an external electric field gradient instead of an optical pulse was proposed [34–36]. The idea is to use optical pumping to create a narrow spectral absorption window, which is then broadened by an external field - the scheme was named *controlled reversible inhomogeneous broadening* (CRIB). A modified version of the CRIB protocol was later developed, known as *gradient-echo memory* (GEM).

Later, the *atomic frequency comb* (AFC) protocol was proposed [37, 38], based on the creation of a series of narrow, highly absorbing peaks by tailoring the inhomogeneous absorption spectrum of an optical transition $|g\rangle \rightarrow |e\rangle$.
Originally, the AFC protocol was proposed for temporal multimode storage, which proved difficult using the CRIB scheme due to the number of modes scaling with the optical depth

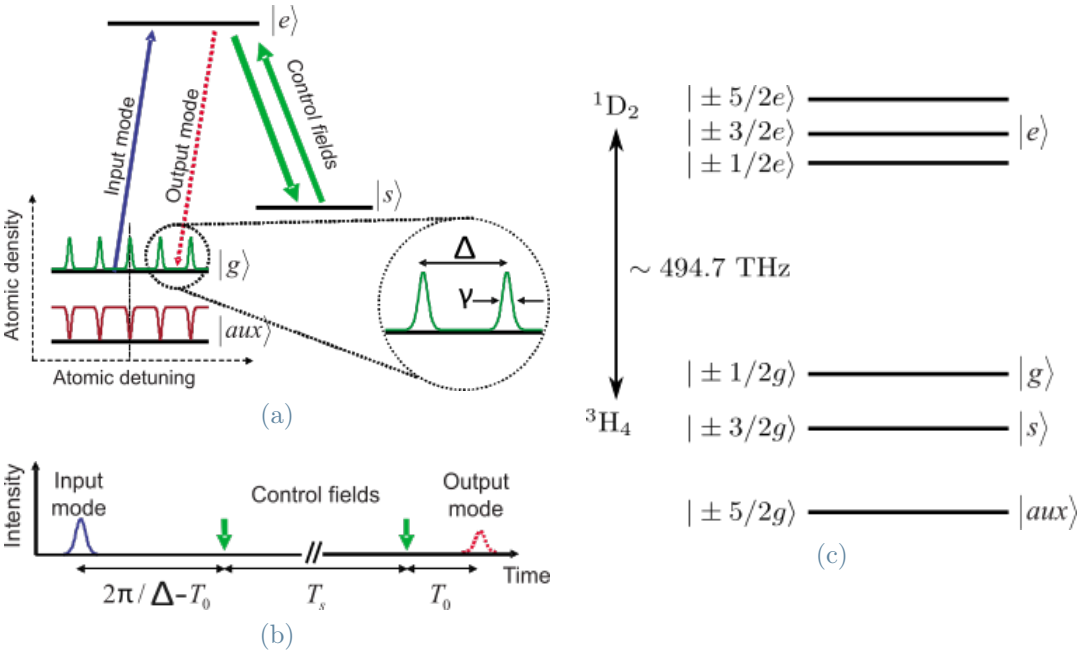


Figure 1.4: AFC-spin wave protocol for quantum memories. a) an inhomogeneously broadened optical transition $|g\rangle \rightarrow |e\rangle$ is shaped into an AFC by frequency-selective optical pumping to the $|aux\rangle$ level. The peaks in the AFC have width γ and are separated by Δ, so that we can define the comb finesse F=Δ/γ. b) The input mode is completely absorbed and coherently excites the AFC modes, which will dephase and then rephase after a time $2\pi/(\Delta - T_0)$, resulting in a photon-echo type coherent emission. A pair of control fields on $|e\rangle \rightarrow |s\rangle$ allows for long-time storage as a collective spin wave in $|s\rangle$, and on-demand readout after a storage time T_s. Images taken from [37]. c) Energy levels of the Pr^{3+} ion inside a Y_5SiO_2 cristal host.

of the material - this does not happen with the AFC protocol, as here the number of modes depends on the number of peaks in the combs, which in turn depends on the ratio of the comb periodicity to the total comb bandwidth. The protocol works as showed in Figure 1.4. Right after the single photon absorption event at time $t = 0$, the atomic state is described by a collective Dicke state

$$|\psi(t=0)\rangle = \sum_{k=1}^{N} c_k \, |g_1 \cdots e_k \cdots g_N\rangle \tag{1.9}$$

where the amplitudes c_k depend on the detuning and position of the particular atom k. Initially, the modes are all in phase, but the system evolves since each terms acquires a

rotating phase term depending on the detuning δ_k of each exctited atom:

$$|\psi(t)\rangle = \sum_{k=1}^{N} e^{-i2\pi\delta_k t} c_k \, |g_1 \cdots e_k \cdots g_N\rangle \, . \tag{1.10}$$

If the peaks are sufficiently sharp, we can approximate the detunings as a discrete set with spacing Δ, such that $\delta_k = m_k\Delta$, where m_k are integers.
From equation 1.10, it follows that the original state is recomposed after a time $\tau = 1/\Delta$, at which moment a coherent photon-echo is re-emitted in the spatial mode defined by the absorbed photon.

In order to achieve longer storage times and on-demand retrieval of the stored photons, the collective excitation can be transferred to an auxiliary long-lived ground state $|s\rangle$ before the re-emission. This way, the evolution of the atomic dipoles is frozen and the excitation is stored as a collective spin wave, which in certain REI-doped crystal has extremely long coherence times. The readout is achieved by transferring back the excitation to $|e\rangle$ after a storing time T_s, where the rephasing of the atomic dipoles happens and the a photon is re-emitted.
In this configuration, the states $|g\rangle$, $|e\rangle$ and $|s\rangle$ form a so-called Λ-system, and the adapted versions of the already described protocols are the Λ-GEM and the Λ-AFC, or *spin wave AFC*.

Both in the normal and spin-wave extended version, an additional auxiliary state $|aux\rangle$ is needed for storing population that has been pumped away during the creation of the AFC. Overall, three ground state levels are needed, plus a coherent spin-transition $|g\rangle - |s\rangle$ must be allowed. This, together with the requirement of a low magnetic dipole moment, strongly limit the choice of the REI to use.

State of the art of integrated quantum memories

We will now review some of the most recent advancements on quantum memories, focusing on realizations of the AFC protocol.

The first realization of the AFC protocol has been demonstrated in 2008 at the University of Geneva [37], where single photon-level pulses were stored in a Nd^{3+}:YVO_4 crystal. After that, rapid improvements both on efficiency and multimodality allowed to demonstrate storage of true single photons [31, 39].

Initially, the AFC protocol was used for the storage of time-bin encoded qubits in rare earth ion-doped crystals [23]. More recently, storage of temporally multiplexed polariza-

tion qubits has been demonstrated [40] also at telecommunication wavelength [41]. Orbital angular momentum (OAM) entanglement has been stored in a Nd^{3+}:YVO_4 quantum memory, demonstrating the spatial multimode capacity of rare earth-ion doped crystal quantum memories [42], adding to the already known temporal [38, 40] and spectral [43] multiplexing capabilities.
The best performances in terms of coherence times have been obtained with spin storage. Heinze et al. [44] demonstrated efficient operation together with long coherence time in a Pr^{3+}:Y_2SiO_5 QM, with storage times up to 1 minute.
Later, Jobez et al. [45] showed cavity-enhanced storage in a $^{151}Eu^{3+}$:Y_2SiO_5 memory, using the AFC-spin wave protocol.
The high-water mark for spin storage is due to Zhong et al. [46], who extended the coherence time of ground-state hyperfine transitions in $^{151}Eu^{3+}$:Y_2SiO_5 to 370 ± 60 minutes at 2K.

While the best performances are obtained in bulk materials, integrated quantum memories have recently gained interest due to the ease of interfacing with other optical circuits. After the seminal work by Corrielli et al. [47], laser-inscribed quantum memories have seen a rapid increase in popularity due to their compactness and scalability.
On-demand retrieval of true single photons has recently been shown for frequency multiplexed photons [48] and polarization encoded qubits [49] in $^{151}Eu^{3+}$:Y_2SiO_5, using the spin wave AFC protocol.
Moreover, storage of temporal-bin encoded qubits within the AFC scheme has recently been shown in a fiber-integrated Pr^{3+}:Y_2SiO_5 QM by Rakonjiac et al. [50].
Laser-written quantum memories will be discussed in deeper detail in Section 2.4.

In this context, this thesis work is aimed at realizing a glass-crystal integrated quantum memory for the storage of photonic entanglement within the AFC scheme.

2 | Femtosecond Laser Micromachining

Femtosecond laser micromachining (FLM) for photonic applications was pioneered by Davis et al. [51] in 1996, when high quality waveguides were inscribed in a fused silica substrate exploiting the nonlinear absorption of an infrared fs laser to create localized refractive index modifications in the glass.
Since then, FLM has proven a solid and versatile way to fabricate photonic circuits in a variety of materials (glass, transparent polymers, crystals [52]) and has been employed for several applications ranging from the fabrication of integrated circuits for quantum photonics [53] to the integration between optical and microfluidic circuits [54, 55].

FLM shows some advantages over traditional lithographic techniques for integrated photonics circuits fabrication.
First, it's a maskless process and requires no clean room facilities, allowing for fast prototyping and realization of small number of devices. Moreover, it allows for 3D writing, which is impossible with lithography - indeed, the nonlinear modification of the material only occurs where the laser is focused, and the focusing depth can be changed without altering the properties of the modified regions [56]. Finally, different substrates can be used for FLM on the same fabrication setup by simply tailoring the fabrication parameters. This is due to the nonlinear nature of the process, the only constraint for FLM writing being the transparency of the substrate. The versatility of this micromachining process extends beyond the fabrication of optical waveguides. For example, it is possible to use FLM to enhance the etchability of glass with respect to a wet etching attack (e.g. by hydrofluoric acid), and exploit this effect to fabricate buried and 3D microchannels. [57].

2.1. Physical principles

FLM is based on the strong nonlinear absorption of a fs laser beam by a transparent material. The nonlinear behaviour is necessary for waveguide writing - this way, the

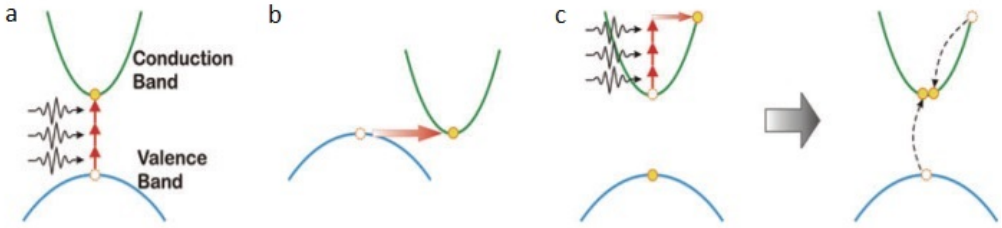


Figure 2.1: Nonlinear photoionization processes at the basis of FLM. a) Multiphoton ionization. b) Tunnel ionization. c) Avalanche ionization. Image taken from [58].

interaction is only limited to the focal volume and the modification of the material is localized.

For this to happen, laser pulses must have high peak intensity - of the order of 10 TW/cm^2 - and energy lower than the energy gap of the substrate. Then, the physical interaction bringing to the permanent index modification can be divided into two steps. First, energy is accumulated in the focal volume, creating ionized electron plasma. Second, energy is released to the lattice and the material is modified.

2.1.1. Plasma formation in the focal volume

In order to enter the nonlinear regime, laser pulses with energy lower than the bandgap of the substrate are used. This way, only multiphoton absorption is possible. In turn, nonlinear processes require high intensities, so that only the tight volume around the focal spot of the laser undergoes any interaction. Then, the laser is able to ionize valence electrons in the conduction band, leading to the formation of plasma.

The ionization of valence electrons can be ascribed to three processes.

Multiphoton absorption

The simultaneous absorption of m photons with energy $h\nu$ such that

$$mh\nu \geq E_g$$

is required to ionize a valence electron, where E_g is the material's bandgap. When low intensities and high energies are employed, phew photons are needed for each absorption, so that this ionization process has high probability.

Tunnelling ionization

On the other hand, at high intensities and low frequencies the field is strong enough to distort the band structure of the material. If the effect is strong enough, the barrier between valence and conduction band is lowered enough to allow ionization by quantum tunnelling between the two bands [1].

Avalanche ionization

Electrons which are already in the conduction band can absorb light by free carrier absorption. If a conduction electron with initial energy E_0 absorbs m photons such that its final energy is $E_{el} = E_0 + mh\nu \geq E_g$, it can ionize a valence electron by impact ionization. Notice how the absorption events don't need to be simultaneous, making this process more probeble than multiphoton absorption. After the first impact ionization, two electrons are found in the conduction band bottom, which can both absorb single photons and repeat the process, giving rise to an avalanche ionization. There are two limiting mechanisms to avalanche ioniaztion [60]:

- Avalanche ionization continues until the plasma becomes highly absorbing, i.e. when the plasma frequency
$$\omega_p = \sqrt{\frac{n_e e^2}{m_e \varepsilon_0}}$$
approaches the laser frequency (n_e being the electron population density).
- A certain number of seed electrons must be present in the conduction band to start the process. These electrons can be provided by the other absorption processes or, if the laser intensity is not high enough, by thermal excitation and thermal defects. The former is usually the case for ps or ns lasers, which produce broader pulses with lower peak intensities [61, 62]. However, avalanche ionization seeded by defects is highly inhomogeneous due to the dependence on the distribution of the impurities. For this reason, fs lasers are used to seed avalanche ionization with absorption-generated electrons and obtain smooth and homogeneous modifications.

[1] The two rival regimes of multiphoton absorption and tunnelling ionization can be described within the same framework by using the Keyldish parameter Keyldish [59]
$$\gamma = \frac{\omega}{e}\sqrt{\frac{m_e c n \varepsilon_0 E_g}{I}}$$
where ω is the laser frequency and I is the focus intensity. Then, if $\gamma \gg 1.5$ - high frequency, low intensity - multiphoton absorption dominates, while if $\gamma \ll 1.5$ - low frequency, high intensity - tunnelling ionization becomes important. In FLM, $\gamma \simeq 1$, so that both processes concurr to plasma formation.

Avalanche ionization makes FLM versatile in the choice of the material. Indeed, multi-photon absorption and tunnel ionization depend in a complex way on laser frequency and material bandgap, therefore their likelihood strongly depends on the material of choice. On the other hand, avalanche ionization depends only linearly on those parameters, hence it can be initiated in very different materials by tuning the laser parameters.

2.1.2. Relaxation process and permanent modification

After the hot electron plasma is generated, its energy is released to the lattice, ultimately causing the permanent modification of the material. Since energy relaxation and heat diffusion happen in times ranging from several picoseconds to few microseconds [63], these processes can be considered as decoupled from plasma formation if the incident laser pulses are in the sub-picosecond regime. This is another reason for choosing fs lasers over ns or ps lasers, since decoupling plasma formation and energy relaxation increases the confinement precision of the material modification.

The physical mechanisms leading to the permanent modification starting from plasma relaxation are not yet fully understood. Still, three different regimes can be identified for silica glasses depending on the fabrication parameters (mainly pulse energy and duration, scanning speed, laser repetition rate) [64].

Smooth refractive index change

At low pulse energies (~100 nJ) a smooth, permanent index modification is observed. Such modification can be attributed to rapid quenching of the melted glass in the focal volume [65], color center formation [66], ion migration effect [67] and pulse-induced shock waves [68]. This regime is the best one for waveguide writing since the material keeps its transparency and optical quality - indeed, smooth and uniform refractive index modifications are required to obtain low propagation losses.

Birefringent refractive index change

For higher pulse energies (~150-500 nJ), birefringent index modifications together with the formation of periodic nanogratings are observed in bulk fused silica [69]. In particular, Hnatovsky et al. obtained subwavelegth period nanogratings with orientation perpendicular to the laser polarization [70]. This second regime with nannogratings formation is unique to fused silica and has not been observed in any other substrate.

Void formation

At even higher pulse energies ($>$ 500 nJ) peak intensities reach $\sim$ 100 TW/cm^2, and shock waves are generated after energy is released from plasma to the lattice. These shock waves leave a hollow core surrounded by a high refractive index cladding. Although not suitabale for light guidance, void structures can be employed for 3D optical memories [71] or photonic bandgap structures [72].

2.2. Waveguide writing

Waveguides are fabricated by translating the substrate under the laser. Since the final guiding properties o the structure are largely depending on the amount of deposited energy in the substrate and the way it is deposited, different fabrication parameters play a role in the final result.

2.2.1. Writing parameters

Focusing

The laser pulses are focused inside the material to achieve a micrometer-sized focal spot and high peak intensities to drive the nonlinear absorption. Usually a microscope objective is used, compensated for both chromatic and spherical aberrations that could reduce the quality of the fabrication. In the approximation of linear propagation in a dielectric medium with refractive index n, the properties of the laser beam can be described by considering its Gaussian profile. The spot size at the focus w_0 and the Rayleigh parameter[2] z_0 are [58]

$$w_0 = M^2 \frac{\lambda}{\pi NA} \tag{2.1}$$

$$z_0 = M^2 \frac{n\lambda}{\pi NA^2} \tag{2.2}$$

where M^2 is the Gaussian beam propagation factor, λ is the laser wavelength and NA is the numerical aperture of the objective. From equations 2.1 and 2.2, it's evident how the numerical aperture plays a key role in defining the shape of the focal volume. Indeed, we have

$$\frac{w_0}{z_0} = \frac{NA}{n}$$

which means that a higher numerical aperture will produce a less elongated focus. However, high aperture objectives are more prone to spherical aberrations induced by diffraction at the air-glass interface. A solution to this problem is to use particular objective

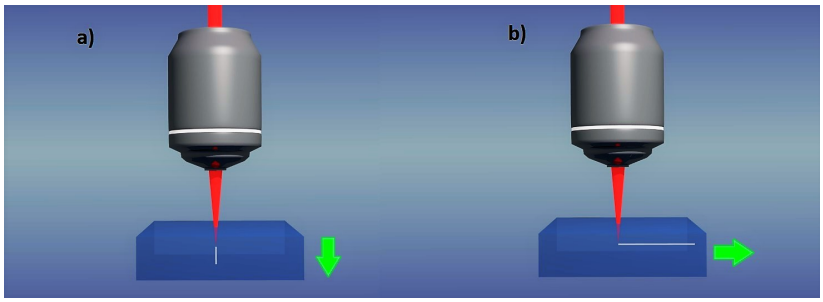


Figure 2.2: Waveguide fabrication geometries. The green arrows represent the sample translation directions. a)Longitudinal writing. b) Transversal writing.

with a correction collar, designed for aberration compensation at a certain depth inside the glass. Else, oil or water immersion gives a smoother environment-glass index contrast at the cost of a lower working distance [73].

Writing Geometry

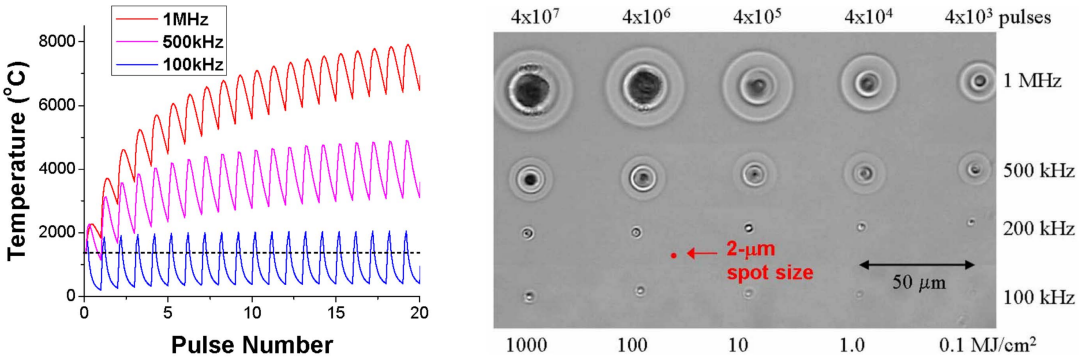
Two writing configurations are possible:

- **Longitudinal writing**: the sample is translated on a direction parallel to the beam. Given the cylindrical symmetry of the transverse profile of the laser beam, waveguides written in this way show cylindrical symmetry as well. Nevertheless, this configuration does not allow to fabricate long waveguides because of the limitations imposed by spherical aberrations - as discussed above - and objective working distance, usually not higher than few millimeters.
- **Transversal writing**: the sample is translated on the plane orthogonal to the laser beam. While the problems related to depth sensitivity are solved, waveguides written this way are highly asymmetrical because of the elongated focal volume. For this reason, waveguides support elliptical modes and show non-negligible birefringence, which is a trait useful for some applications but undesired for others. This effect can be compensated by using beam shaping techniques [74, 75] or by exploiting thermal annealing treatments (see section 2.2.2).

Repetition Rate

The repetition rate of the laser beam is of key importance, since it determines the degree to which energy accumulation and relaxation are coupled.

[2]Starting from the focal spot, distance along the propagation direction of the beam where the beam's cross section is double the spot size.



(a) (b)

Figure 2.3: a) Numerical simulations of glass temperature versus exposure time at different repetition rates: low (blue), intermediate (magenta), high (red). b) Microscope images showing the increase in dimensions of the modified regions for growing repetition rates (rows) and exposure (columns). Images taken from [76].

For low repetition rates (< 250 kHz), the interval between two subsequent femtosecond pulses is longer than the typical time scale of the relaxation processes, of the order of few μs. Therefore, after one pulse heats up the glass a complete heat relaxation takes place before the next pulse arrives. Tow major drawbacks arise. First, since the modification is mainly due to individual pulses, which means the waveguides have strongly asymmetrical transversal sections, leading to birefringence and significant coupling losses. Second, low translation speeds are required to deposit enough energy in the dielectric, leading to long fabrication times.

At high repetition rates (several MHz) the interval between two subsequent pulses is smaller than heat diffusion processes. The effect of pulses adds up and a strong heat accumulation in the focal volume takes place. This way, waveguides with larger diameter and a more uniform section can be obtained, showing lower losses. With high repetition rates, higher translation speeds can be used, lowering the fabrication time. However, since pulse energy is usually smaller, tight focussing with a high numerical aperture objective is required to reach high intensities.

In an intermediate region - between few hundred kHz and few MHz - the advantages of the two regimes combine. Pulse energy is still rather high, removing the requirement for tight focussing, and heat accumulation due to subsequent pulses still occurs, leading to more symmetric waveguides with a diameter comparable to the spot size of the laser. This regime is widely used for waveguide fabrication.

Other parameters

Other fabrication parameters influence the guiding properties of laser written waveguides:

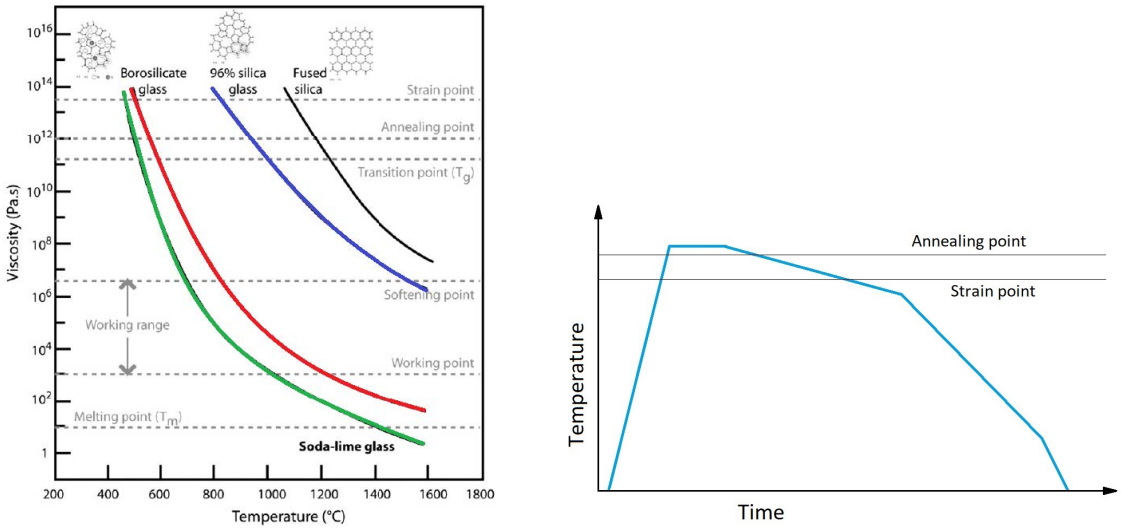
- Translational speed influences the rate at which energy is deposited in the sample.
- Laser wavenelgth influences the order of the multiphoton absorption process. For some materials, lower inscription wavelengths are preferred [77].
- Polarization of the laser beam influences the guiding properties of structures in some materials, e.g. fused silica [78].
- The writing direction affects the guide profile because of the so-called "quill effect", due to phase front tilting of the laser beam [79].

2.2.2. Waveguides in glass

While waveguide writing is in pricniple possible in every glass material, few glasses are actually used for photonic applications.
Since the first demonstration of waveguide inscription in fused silica, further demonstrations of high-contrast waveguides were proposed. In 2011, Eaton et al. [80] were able to fabricate waveguides with an index contrast of 0.022 and mode size as low as 7 µm for light at 1550 nm, by using 0.55-NA oil immersion focusing of a 522 nm femtosecond laser with repetition rate of 500 kHz onto a fused silica substrate. The use of a high repetition rate allows to use higher scanning speeds, greatly reducing the fabrication times with respect to previous works [81].
While fused silica retains its importance in applications such as optofluidics due to the possibility to apply etching techniques [82], boro-aluminosilicate glass is the most widely used glass for non-classical photonics due to its large and uniform transmission window.
Thermal processes are usually employed as a post-fabrication step to obtain smoother index profiles and lower mode diameters.

Thermal annealing

As discussed in the previous sections, very high temperatures are reached inside the material during fabrication, even beyond the melting point. Fast solidification - necessary to obtain a vitreous modification, not crystalline - leads to accumulation of mechanical stress, which can be reduced by employing thermal treatments.



(a) (b)

Figure 2.4: a) Viscosity of different types of glass. The characteristic temperatures are highlighted. Image taken from [83]. b) Temperature vs time diagram for a typical glass annealing process.

To understand how glass annealing works, we introduce the characteristic temperatures of glass, defined based on its viscosity:

- **Strain point**: temperature corresponding to a viscosity of $10^{14.5}$ Poise ($10^{13.5}$ Pa·s). At this temperature, glass is solid and keeps its shape, internal stresses are released in a few hours.
- **Annealing point**: temperature corresponding to a viscosity of $10^{13.4}$ Poise ($10^{12.4}$ Pa·s). At this temperature, stresses are released in a few minutes by viscous relaxation.
- **Softening point**: temperature corresponding to a viscosity of $10^{7.65}$ Poise ($10^{6.65}$ Pa·s). At this temperature, glass is soft enough to deform under its own weight.
- **Working point**: temperature corresponding to a viscosity of 10^{4} Poise (10^{3} Pa·s). At this temperature, glass is sufficiently soft to be deform by blowing or pressure.
- **Melting point**: Since a phase transition does not occur in glass, being it amourphous, a proper melting point cannot be defined. Depending on the composition of the glass, a melting point can be identified to distinguish soft glass from liquid.

The annealing process exploits the different properties of glass at different temperatures. First, the glass is heated slightly above the annealing point. Temperature is kept constant for a few minutes to allow internal stresses to relieve.

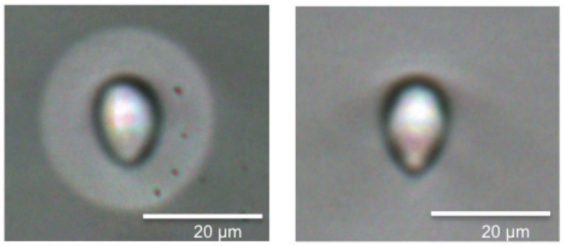


Figure 2.5: Waveguide profile before (a) and after (b) the annealing process.

Then, temperature is reduced below the strain point at a slow rate - few °C/h - to allow uniform cooling of the glass.
Once below the strain point, glass can be cooled down with a higher rate until room temperature is reached.

The effect of annealing on guiding properties of laser written waveguides were first studied by Arriola et al. in 2013 [84]. They proposed a two-step fabrication process to obtain single-mode waveguides at 1550 nm with low losses, written in boro-aluminosilicate glass (EAGLE 2000 by Corning).
First, multimode waveguides are written using low transition speed and high pulse energy (500 mm/min and 90 nJ). This way, a relevant modification to the substrate is caused. Microscope imaging of the waveguide section reveals a solidified core surrounded by a outer ring modification.
After the fabrication, an annealing procedure brings to stress relaxation, and in the end the outer modification ring is removed. Moreover, the index profile of the core is now Gaussian-like and its diameter is reduced. In the same work, the authors also suggested how the annealing process would reduce the birefringence of the waveguides, since it mainly arises from stress accumulation in the material.

2.2.3. Waveguides in crystals

Optical dielectric crystals are important for a variety of applications, such as electro optic modulators [85], frequency converters [86, 87], waveguide lasers [88] or optical quantum memories [47], which are the aim application of this work.

The versatility of FLM allows to fabricate different guiding structure, depending on the application. The key concept is to construct a guiding structure composed of a higher refractive index core - the waveguiding core - surrounded by lower index cladding layers, which could be modified or unmodified substrate regions or air.
While the physical mechanisms leading to material modification are still the same de-

scribed in the previous sections, the high positional order of crystals makes it difficult to achieve smooth index modifications in certain cases. Indeed, it has been shown that both a positive or negative refractive index change can be induced in the focal volume by laser inscription, depending on the material and fabrication parameters.

Laser micromachined waveguides in crystals are generally classified into four groups based on the type of modification induced [89, 90].

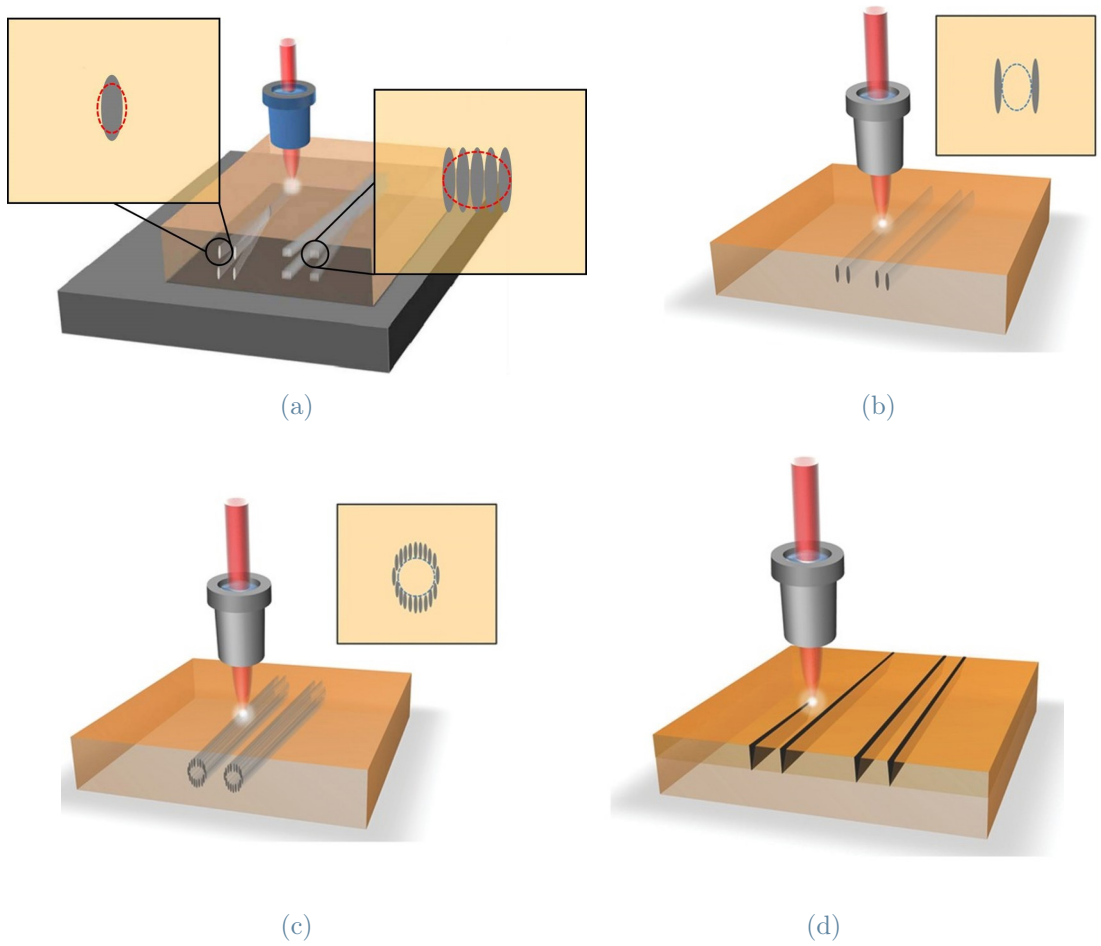


Figure 2.6: Schematic of the fabrication process of fs inscribed waveguides in dielectric crystals. a) Type I single-scan (left) and multi-scan (right). b) Type II. c) Type III. d) Type IV. Images adapted from [89, 91].

Type I waveguides

In this configuration, the fs laser induces positive refractive index variations in the focal volume, which constitute the waveguide core (Figure 2.6a). The waveguide can be constituted by a single track (*single line* waveguides) or by multiple laser scans overlapped

transversally to the writing direction (*multiple line* waveguides) [92]. This is done by irradiating the substrate with a low-intensity laser, below the damage threshold. While this is the usual way to write waveguides in glass, such a modification is more difficult to obtain in crystals due to their periodic structure and has only been shown in few materials.
Type I waveguides have three main drawbacks [91]:

- Due to the combined effect of waveguide birefringence with the often occurring natural birefringence of the crystal, type I waveguide often support the light guidance of only one polarization mode.
- The distinguishing properties of crystals arise from their periodic structure. Index modifications come as a consequence of a high damage - or even amorphization - induced by the laser in the focal volume, which means that bulk-related properties such as coherence or nonlinear properties can result degraded in the waveguide core.
- Single line waveguides in crystals are usually completely removed under high temperature, hampering their use for high power applications.

While the number of applications of Type I waveguides is strongly reduced by the polarization sensitivity and temperature restraints, they can still be used to produce passive photonic circuits (e.g. directional couplers) provided the bulk properties are preserved to a certain degree.

Type II waveguides

In this configuration, the guiding structure is in the vicinities of the fs-induced modification. As shown in Figure 2.6b, a negative index variation is induced in the irradiated region by dilation of the focal volume. The region near the core will posses a higher index due to stress-induced effects. Usually, two parallel modification tracks are inscribed, so that the region in-between serves as the waveguide core (*double line* geometry).

With respect to Type I waveguides, the bulk features of the crystal are mostly preserved in the waveguide core, since the modification only affects the surroundings [93]. Further, the refractive index changes in the irradiated tracks are more easily controlled than in Type I waveguides. Moreover, it's possible to engineer the track distance to support guidance for both polarization modes in the waveguide region. Finally, we mention how such stress-induced waveguides are stable at high temperatures, making suitable devices for high power applications. It has also been shown how guiding properties can be further improved by employing more than two damage tracks (*multiple-line* geometry) [94].

Type III waveguides

Similar to Type II waveguides, the waveguide core is in the stress-induced high index region between low-index irradiated tracks. The difference is that in this case the irradiated tracks form a quasi-closed geometry around the core (Figure 2.6c), which makes for a quasi-continuous low-index potential well (*depressed cladding* waveguides).

One key feature of Type II waveguides is that the waveguide core is far from the modification tracks, preserving the bulk properties of the crystal even better than in Type II guides. Further, the diameter of the track arrangement can be modified to enable guidance from single to multimode and from visible to mid IR light, and to allow optimal coupling with fibers or other integrated devices. The geometry flexibility also allows for almost identical guidance along the two polarization modes. Type III waveguides have recently gained popularity, for example as a way to integrate photonics devices with crystals [95] or to improve the performance of waveguide lasers [96].
Still, their large size and fabrication complexity pose important constraints to the realization of curved and/or branched structures.

Type IV waveguides

In this condiguration, the laser is used to etch the crystal in selected regions by ultrafast ablation. Figure 2.6d shows the schematic of a Type IV waveguide, also called *ablated ridge* waveguide. Two tracks are ablated from a planar waveguide layer, to form a potential well that allows light guidance in the region in between the tracks. While the concept is similar to that of Type II waveguides, this configuration allows to fabricate guiding structures on multi-layer crystals in a simple way [97]. Still, one major drawback lies in the roughness of the etched tracks, which introduces losses and degrades the waveguide quality - this can be improved by employing post-fabrication treatments [98].

2.3. State of the art of integrated passive photonic devices

2.3.1. Reconfigurable interferometric circuits in glass

Interferometers are core devices for photonics applications. They enable direct control of the interference between multiple modes, allowing to perform complex operations in integrated devices giving compact dimensions and high scalability and robustness.

One key property of integrated photonic devices is reconfigurability. For instance, it can

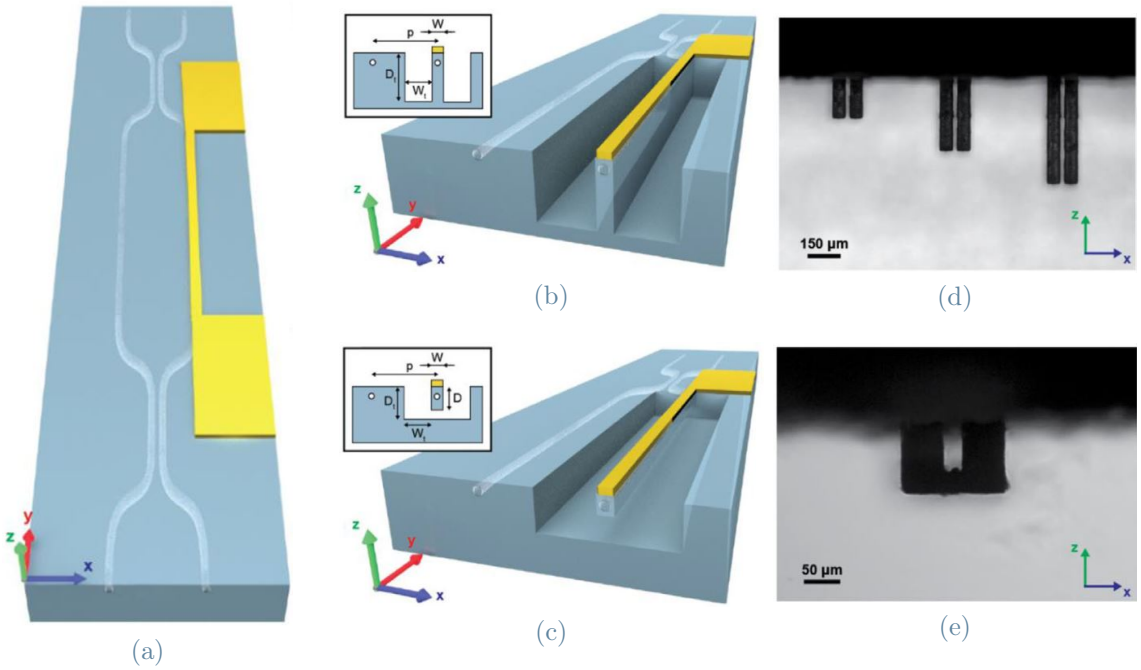


Figure 2.7: The reconfigurable MZIs described in [100]. a) Basic structure of the device with no isolation structures. b) Cross section of the device with deep isolation trenches. c) Cross section of the device with bridge isolation. d) Microscope image of the isolation trenches of different depths seen through the side of the substrate. e) Microscope image of the bridge waveguide seen through the side of the substrate. Images taken from [100].

be used to compensate for errors due to fabrication tolerance, or to reconfigure the transfer matrix of a device, allowing to perform different experiments with a single chip.
Reconfgurability is obtained by using thermal shifters, which exploit the dependency on temperature of the refractive index. Experimentally, thermal phase shifters are now obtained by patterning thin stripes of metal deposited on top of the substrate.
Here, we review the key technological advancements for the technologies, together with four experiments that show its significance.

Flamini et al. [99] first inscribed a Mach Zehnder interferometer by FLM in a borosolicate substrate, with a thermal phase shifter on top of one arm. The resistor was patterned on a gold film, using the same femtosecond laser employed for the inscription of the interferometer. Fringe visibility higher than 95% was shown in experiment with single- and two-photon states.

One major problem of this simple configuration is the scalabiliy of the technique. Indeed, thermal crosstalk hinders the use of thermal phase shifters in more complex circuits, where phase control must be precise and heating must be localized.

A solution to this obstacle was proposed by Ceccarelli et al. [100], who showed how the use of 3D isolating microstructures improves both thermal crosstalk and power dissipation by orders of magnitude, allowing to concentrate heat dissipation around the waveguide on which the phase shift is desired. Two different microstructures were described, obtained by etching the substrate around the waveguide by laser ablation.
First, *trench* microstructures are fabricated by removing glass boxes from both sides of the microheater. The boxes are removed as close as possble to the waveguide, to limit the amount of glass subject to heating and reduce the width of the glass slab connecting the waveguide to the rest of the substrate. Second, *bridge* microstructures are based on the previous ones, only this time glass is removed also under the optical path, encasing the waveguide portion under the microheater in a suspended bridge. The power dissipation needed to induce a full 2π shift was as low as 37 mW, and the crosstalk reduced to a few percent points.
In the same work, the authors proposed a novel method to improve the fabrication process of thermo-optic phase shifters. First, the gold layer is deposited after a thin Chromium buffer layer, which improves the Gold-glass adhesion. Then, an annealing process is tailored to improve the film stability and ensure stable performances of the resistors over time. First, temperature is raised to 400 °C with a ramp of 10 °C/min, then it's kept stable for 30 min, and finally the sample is brought to $\sim$ 20 °C in 15 h. Being the maximum temperature of the process far from the strain temperature of glass, the substrate remains unaffected by the procedure.

Parallel to the work of [99], Chaboyer et al. [101] realized a tunable three arm interferometer by directly mounting a heater on top of the glass sample. To do so, an array of thick-film aluminum stripes was glued to the glass sample with a thermal paste. Figure 2.8 shows the schematics of the circuit.
The device was used to demonstrate control of two-photon interference, showing stable and reproducible phase measurements below the standard quantum limit. This work pioneered the application of integrated interferometric devices to quantum metrology and enhanced multi-parameter extimation experiments. Lately, Polino et al. [102] demonstrated quantum-enhanced performances in multiphase extimation by using two-photon input states, with post-selection of the detection states.
They used a three-port interferometer composed by two cascaded tritters, including six thermo-optic phase shifters implemented with the procedure described in [99], showed in Figure 2.9. In their work, they demonstrated how the use of multiple phase shifters allows to investigate a broader number of parameters, allowing to increase the complexity of the multiparameter extimation problem.

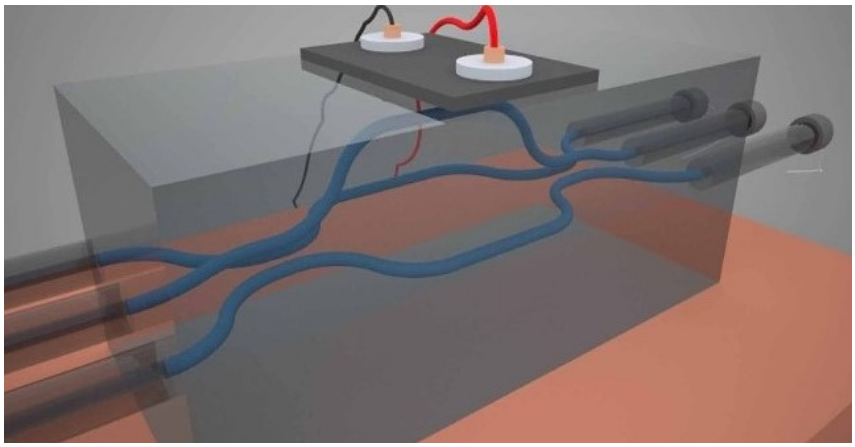


Figure 2.8: Tunable three-arm interferometer used in [101]. One arm is raised with respect to the others, imparting a relative phase due to the interaction with a heater placed on the sample surface. Image taken from [101].

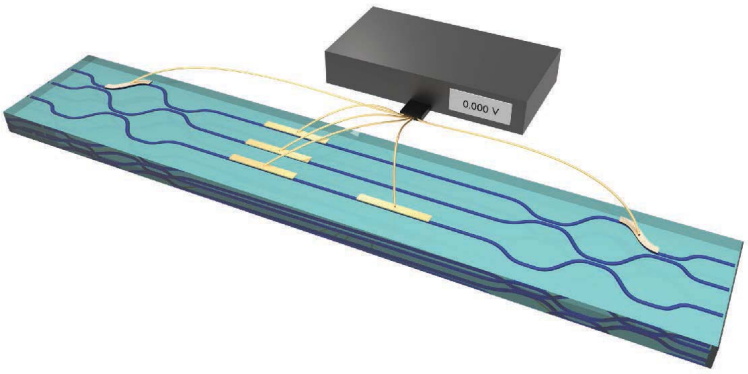


Figure 2.9: Schematics of the reconfigurable integrated three-port interferometer used in [102]. Three straight waveguide segments are included between two multiport splitters. The dynamical control of the phases is achieved by thermo-optic phase shifters. Image taken from [102].

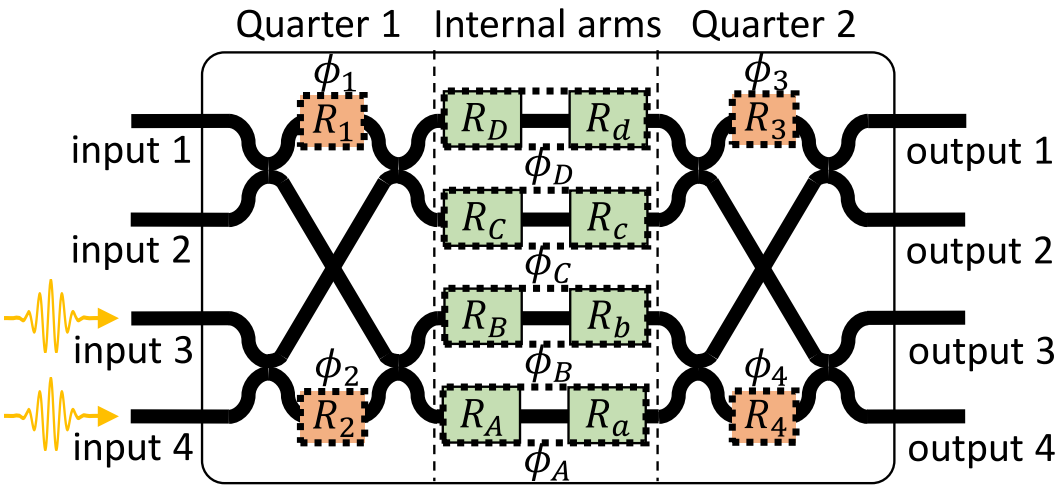
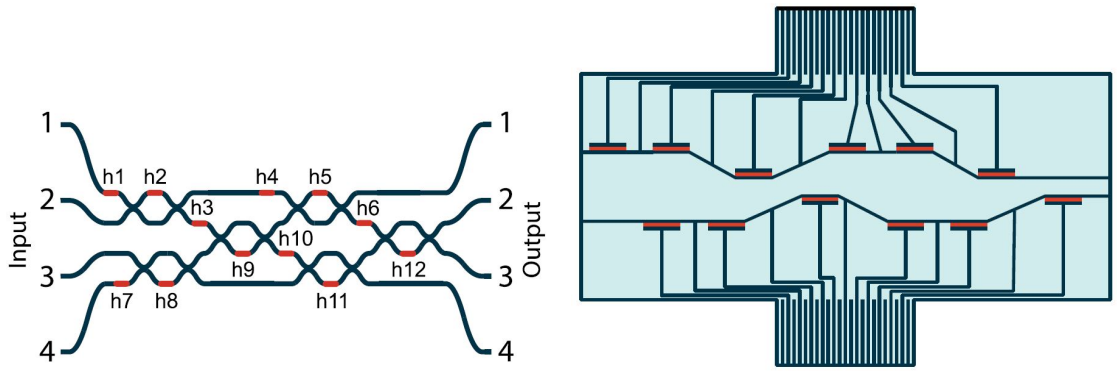


Figure 2.10: Schematics of the reconfigurable interferometer used in [103]. the device is composed of two cascaded quarters, which are 4×4 optical elements that split equally the optical power at all its input ports across all output ports. Each quarter is equipped with two thermal phase shifters to actively control the internal optical phases. Between the two quarters, the interferometric region is composed by four straight waveguide segments whose optical phases can be controlled by means of 8 thermal phase shifters. All thermal shifters have been fabricated according to [104] and include laser ablated isolation trenches around each microheater. This configuration allows to both reduce the power consumption and to greatly reduce the thermal cross talk between adjacent shifters. Image adapted from [103].

Building upon this idea, Valeri et al. [103] have recently proposed a four-port reconfigurable interfeometer to perform quantum-enhanced metrology with two-photon entangled states in adaptive regime, adopting a Bayesian adaptive multiparameter technique together with a control feedback to provide high-performance computation capabilities.
The use of adaptive strategies allows to optimize convergence to the standard quantum limit (SQL) using limited resources, essential for scaling the problem to higher dimensions.
The device they used is described in Figure 2.10. Through the use of a four arm interferometer equipped with a total of ten phase shifters, the authors were able to optimize the simultaneous evaluation of three optical phases.

Reconfigurable devices are of paramount importance also for linear-optical quantum computing and quantum information.
Recently, Dyakonov et al. [105] realized a universally programmable linear-optical circuit, using a 4×4 multiport interferometer equipped with a total of 12 thermo-optical phase

shifters. The device layout is shown in figure 2.11.



(a) (b)

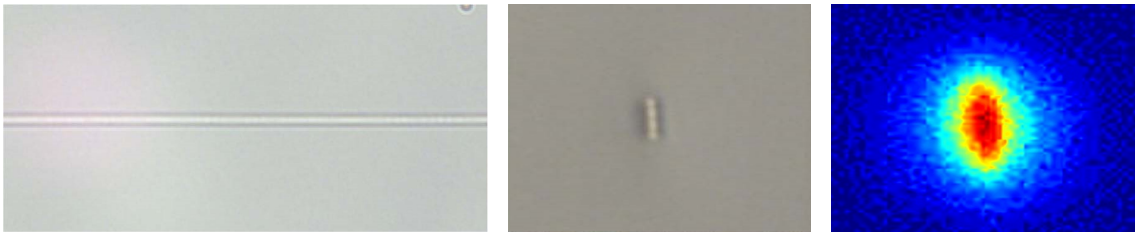
Figure 2.11: Layout of the reconfigurable device of reference [105]. a) Schematics of the interferometric circuit. The thermo-optic phase shifters positions are highlighted in red. b) Layout of the electrical circuitry controlling the phase shifters.

The device was able to realize a universal special unitary group of grade 4 transformation, the highest reported in literature for a FLM inscribed device. They also improved the switching times of the phase shifters, and devised an adaptive algorithm to tune the circuit for the desired output.

2.4. State of the art of laser-written quantum memories

Integrated quantum memories show advantages over bulk devices in term of light-matter interaction intensity, scalability and integrability with other optical devices. Furthermore, rare earth-doped crystals are a suitable candidate for FLM inscription of waveguides. Quantum memory protocols have been demonstrated using waveguide structures of every kind described in Section 2.2.3, inscribed inside suitable crystals.
We will briefly review the advancements and state of the art of waveguide quantum memories in rare eart-doped crystals.

The first realization of a FLM written quantum mermory is due to Corrielli et al. [47]. They fabricated type II waveguides in a Pr^{3+}:Y_2SiO_5 crystal, demonstrating how the laser inscription does not alter the coherence properties of the material and that light confinement in the waveguide increases the interaction with the active ions. Furthermore, they were able to operate the optical protocols needed to prepare the atomic population in



(a) (b) (c)

Figure 2.12: Profile of the type I waveguides used in [107]. a) Longitudinal profile. b) Transversal section. c) Mode profile. Images taken from [107].

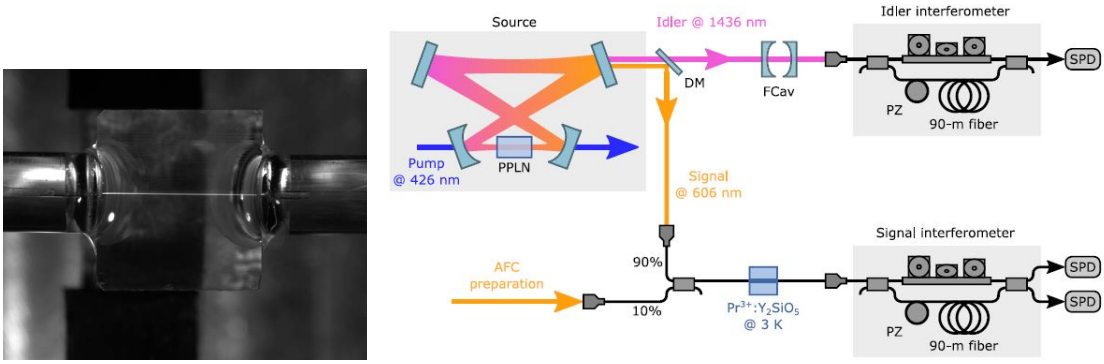
atomic frequency combs, analogous to the bulk crystals, providing a proof-of-principle experiment of light storage using the AFC protocol.
Seri et al. [106] later reported the first single-photon storage in laser written waveguides. They incribed type I waveguides in a Pr^{3+}:Y_2SiO_5 crystal to obtain smaller modes and enhance the light-ion interaction, reporting improved storage times with respect to previous works. The same authors later used the same device for the storage of a heralded frequency-multiplexed single photon using the AFC protocol, containing 15 discrete frequency modes separated by 261 MHz and spanning across 4 GHz [107].

Exploiting the good mode matching between type I waveguides and commercial single mode fibers, Rakonjac et al [50] demonstrated a fiber-integrated quantum memory entangled with a photon at telecomunication wavelength. They used type I waveguides written in Pr^{3+}:Y_2SiO_5, showing improved storage times and efficiency.

While on-demand readout in integrated Pr^{3+}:Y_2SiO_5 systems has only been demonstrated with classical light [47, 107], on-demand operation with quantum light has been shown in $^{151}Eu^{3+}$:Y_2SiO_5 with different types of waveguides.

Zhu et al. [108] first studied the coherence properties of type IV waveguides written in $^{151}Eu^{3+}$:Y_2SiO_5, demonstrating the feasibility of spin-wave storage of quantum light within the AFC protocol.
Later, on-demand retrieval of single photon-level coherent light was demonstrated by Liu et al. [109] in Type IV waveguides written in a $^{151}Eu^{3+}$:Y_2SiO_5 crystal. The authors were able to perform on-demand retrieval of time-bin qubits using the Stark-modulated AFC protocol with high fidelity.

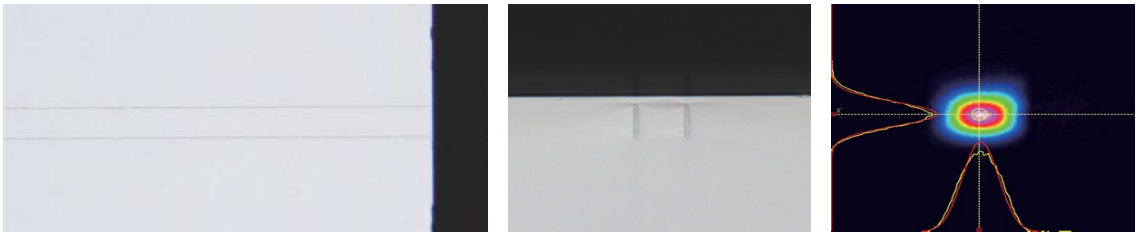
On-demand retrieval of true single photon light was first demonstrated in Type II waveguides by the same authors in [110]. Here, they demonstrate on-demand retrieval of quantum light folliwing both the spin wave AFC and ROSE protocols.
Later, Su et al. [48] demonstrated on-demand storage of frequency multiplexed photons in



(a) (b)

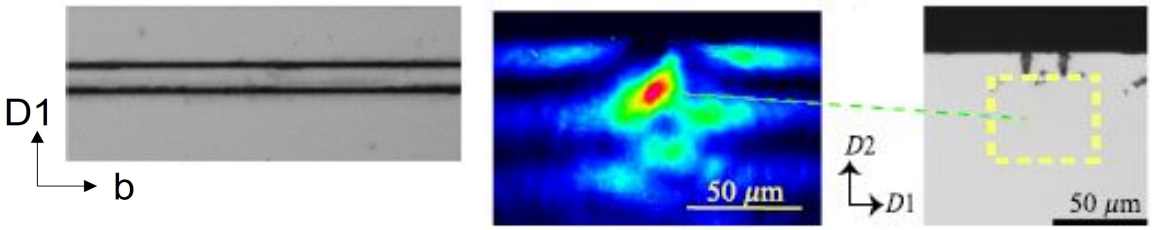
Figure 2.13: Device and experimental setup used in [50]. a) Picture of the FLM inscribed crystal, pigtailed to single mode fibers on the two sides. b) Schematics of the setup. Entangled photon pairs are generated in a cavity-enhanced spontaneous parametric down-conversion (SPDC) source. The idler and the signal are separated by a dichroic mirror (DM), and the latter is coupled to the single-mode fiber that is directly glued to the waveguide in the Pr^{3+}:Y_2SiO_5 crystal, placed inside a cryostat. An AFC of varying storage time is prepared in the memory. The entangled state is analyzed using two fiber-based unbalanced Mach-Zehnder interferometers, with the one for the signal photons directly linked to the waveguide. Images taken from [50].

the same device of [110] following the spin wave AFC scheme. Furthermore, they report on benchmark-low propagation losses, obtained by annealing the waveguide structures after the fabrication. Type III waveguides have recently been investigated as well.
Zhu et al. [49] demonstrated on-demand retrieval of polarization qubits in depressed-cladding FLMW waveguides in a $^{151}Eu^{3+}$:Y_2SiO_5 crystal using the Stark-modulated AFC protocol. In the same work, they demonstrated the frequency-multiplexing capabilities of the device.
Liu et al. [111] demonstrated on-demand storage of single-photon-level light at telecom wavelength in type III FLMW waveguides in a $^{151}Eu^{3+}$:Y_2SiO_5 crystal, implementing the Stark-modulated AFC scheme. Moreover, they pigtailed both the input and output of the crystal to single mode fibers, reporting high fiber-to-fiber transmission.



(a) (b) (c)

Figure 2.14: Profile of the type II waveguides used in [48]. a) Longitudinal profile. b) Transversal section. c) Mode profile. Images taken from [48].



(a) (b)

Figure 2.15: Profile of the type IV waveguides used in [49]. The polarization axis of the crystal are indicated. a) Longitudinal profile. b)Mode profile and transverse section. Images taken from [49].

3 | Experimental Methods

In this Chapter we describe the experimental apparatus used for the fabrication and characterization of our devices.

3.1. Device fabrication

3.1.1. Fabrication setup

The fabrication setup can be divided in three blocks: laser source, beam steering and manipulation, and motion stage, through which the sample is translated under the focused laser. The laser beam first goes through an attenuation stage made of two rotating $\lambda/2$ plates and a polarizing beam-splitter. The first waveplate is actuated by a miniature gear-driven rotary stage (AEROTECH MPS75GR), while the second one is manually controlled.

Using mirrors, the laser beam is lifted from the optical table and focused using a microscope objective. In our experiments, we used a 0.5-NA 20× water immersion objective (Zeiss). The glass sample is fixed to a custom-made holder, filled with deionized water and mounted on a gimbal adjuster.

Alignment of the sample with the laser beam is attained with the aid of a CCD camera (Edmund Optics) that collects back-reflected light from the sample. Both the sample and

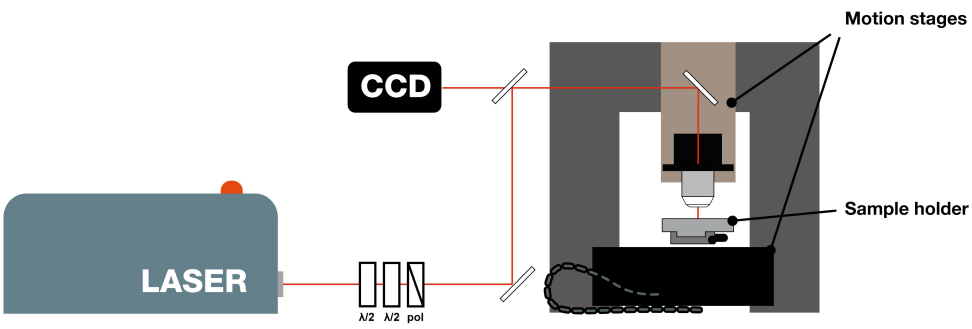


Figure 3.1: General scheme of the fabrication setup.

the objective are mounted on translational stages for precise control of their motion.
A powermeter (Ophir Nova II) can be inserted in the line to measure the beam power.

Laser source

The laser source used is a compact commercial femtosecond laser system (PHAROS by Light Conversion), consisting of a Kerr lens mode-locked oscillator, a regenerative amplifier and a stretcher/compressor module.
The oscillator produces a 76 MHz pulse train, with pulse duration of 80 fs. The gain medium is a Yb:KGW crystal with direct diode-laser pumping. The laser cavity is encapsulated in a monolithic aluminium block to ensure performance stability.

Then, pulse amplification by chirped-pulse amplification (CPA) allows to obtain fs pulses with high peak intensity. The stretcher/compressor in the CPA line is based on a single diffraction grating, adjusted to introduce positive or negative chirp to tune the pulse duration. The regenerative amplifier is based on a diode-pumped Yb:KGW crystal.

The system produces sub-190fs pulses at 1028 ± 5 nm with a maximum pulse energy of 0.2 mJ. The repetition rate can be tuned from single-shot to 1 MHz.

Motion stage

The sample is translated under the laser beam by making use of three independent linear translators moving in the three directions *x, y, z*. The motion stage on the z axis (AEROTECH ABL1500Z-050) is used to move the objective, allowing to tune the focus depth inside the substrate. This stage has a travel range of 50 mm with nominal resolution of 1 nm and maximum speed and acceleration of 300 mm/s and 20 m/s^2. In the plane perpendicular to the laser beam, the sample is translated by two stages - AEROTECH ABL1500WB-200 for the x direction, and AEROTECH ABL1500-100 for the y direction, with maximum travel ranges of 100 mm and 200 mm respectively. Maximum translation speed and acceleration are identical at 2 m/s and 20 m/s^2.

The stages are controlled by a computer equipped with a proprietary software (AEROTECH CNC Operator Interface), which allows to implement G-CODE scripts to program the motion. All the stage drivers feature a TTL output that allows to synchronize the beam shutter to the motion.

3.1.2. Post-processing steps

Annealing process

After the device is fabricated, the glass chip undergoes an annealing process as described in 2.2.2. A commercial oven (Nabertherm LT 9-11-SKM) is used.

Glass polishing

The quality of light coupling in FLM written waveguides strongly depends on the quality of the interface facet. Right after the fabrication, the facet is often chipped, rough or presents surface defects.
To enhance the coupling efficiency, the chip facets are polished by means of moving abrasive disks. The sample is mounted perpendicular to the moving disk and polished using a sequence of disks with decreasing roughness, ultimately reaching optical-grade polishing.

Gold coating and resistor patterning

Thermao-optic phase shifters are fabricated following the procedure illustrated in [104] (see also Section 2.7). Both films of chromium and gold are deposited at the PoliFAB facilities, using a magnetron sputtering system (Leybold LH Z400). After the films are deposited, the sample undergoes another annealing cycle, and then resistors are patterned by laser ablation using the same FLM system used for waveguide inscription.
To allow a precise removla of the metallic film, an 0.25-NA 10× microscope objective (Leitz-Wetzlar) is used, with irradiation power 200 mW and translation speed 2 mm·s^{-1}. To increase the ablation width and improve the isolation between conductive tracks, the contours are ablated 3 times with an offset of 1 m between each inscription.
In order to align the thin ablated resistive stripes with the optical circuit, markers are fabricated by ablating the surface of the glass chip. Then, back-reflection of the laser beam on a CCD camera allows to retrieve the position of the marks, so that the glass chip can be align to the reference frame in different fabrication steps.
Since conventional soldering techniques such as direct wire bonding would damage the thin metal film, electrical connections are realized by direct red tin soldering.

3.2. Characterization setup

3.2.1. Microscope inspection

A first characterization of the device after the fabrication is done with the help of a microscope.
A top view of the chip allows to check the geometry of the fabricated circuit, detecting possible damages or interruptions in the irradiated tracks. Such interruptions may cause losses or completely hinder the guiding properties of the structure. Usually, they are due to the presence of dust or debris on the sample surface during fabrication, that distort the laser focus.
Inspecting the lateral facets allows to check both the quality and roughness of the facet and the waveguide cross-section. This is particularly useful after the polishing step, to check its efficacy, or to compare the cross-sections of waveguides written with different irradiation parameters.
Microscope characterization is performed with a Nikon ECLIPSE ME600 microscope equipped with a PixeLINK PL-B871 CCD camera for image acquisition. Four different objectives are mounted on the microscope - Nikon Plan Flour 4×, 10×, 20× and 40×.

3.2.2. Coupling with classical light

Coupling with classical light allows to investigate the guiding properties of the waveguides, namely its optical losses, supported mode profile, and polarization behavior.
For this characterization, coherent light at 607 nm produced by a diode laser is used. Then, two characterization setups can be used, as shown in figure.

In the *end fire* configuration, the laser beam is focused on the sample input facet by means of an aspheric lens. To reach optimal coupling, both the numerical aperture and focal spot size of the objective should match the waveguide aperture (for FLM-written waveguides, NA~0.1) and mode size. At the output of the sample, light is collected by another objective and imaged on a sensor, which can be a powermeter (Ophir Nova II) or a CCD camera (CCD camera Emdund OpticsEO-1312M).
The second configuration is the *fiber-butt* coupling. Here, the laser light is first coupled to a single-mode fiber (Thorlabs SP-405 XP) whose end, after proper cleaving and peeling, is brought close to the input facet of the sample. The output stage is identical to the previous configuration.

The positioning of the sample and input and collection optics is controlled using a submicrometric positioning system. The sample is mounted on a five-axis micropositioner,

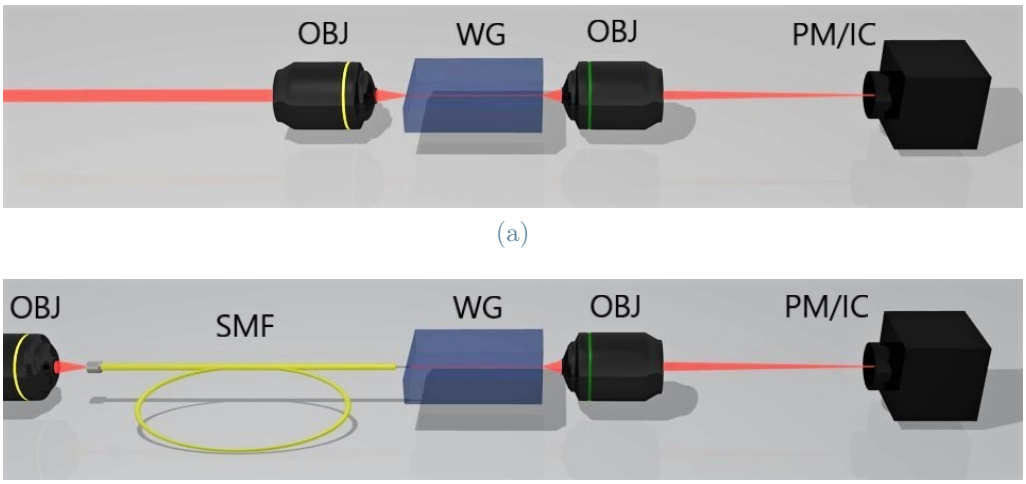


Figure 3.2: a) Setup for *end-fire* characterization. A microscope objective (OBJ) focuses the laser beam on the glass sample with the waveguides (WG), while another objective collects the output light to focus it on either a powermeter (PM) or an imaging camera (IC). b) *Fiber-butt* configuration. This time, the first objective is used to focus the laser beam into a single-mode fiber (SMF), which is then brought close to the input facet of the sample.

while the fiber and collection optics are mounted on two six-axis hexapods (PI H-811). All positioners are computer-controlled, allowing precise control of the position down to 100 nm precision, particularly useful for the characterization of large integrated devices with regualr pitch between waveguides or to perform fiber pigtailing. An optical zoom camera (Opto 100-ZOS7-25) is used to control the sample position and the distance betweeen fiber end and input facet.

Mode profiling

The analysis of the output mode profile allows to extimate the confimenet capabilities of the waveguides and estimate the coupling losses. Moreover, it allows to distinguish between single and multimode waveguides, key distinction in many applications.
In single mode waveguides, the electric field distribution $\mathbf{E}(x,y)$ can be retrieved directly from the intensity profile I(x,y), since $\mathrm{mod}\ \mathbf{E}(x,y) = C\sqrt{I(x,y)}$. Namely, theory shows how sign inversion is absent for the fundamental mode, plus the constant C can be neglected provided the mode profile is nomalized.

Mode profiling is performed in the *fiber-butt* configuration. First, the fiber is coupled to the waveguide, whose output intensity profile is imaged with a CCD camera - taking care

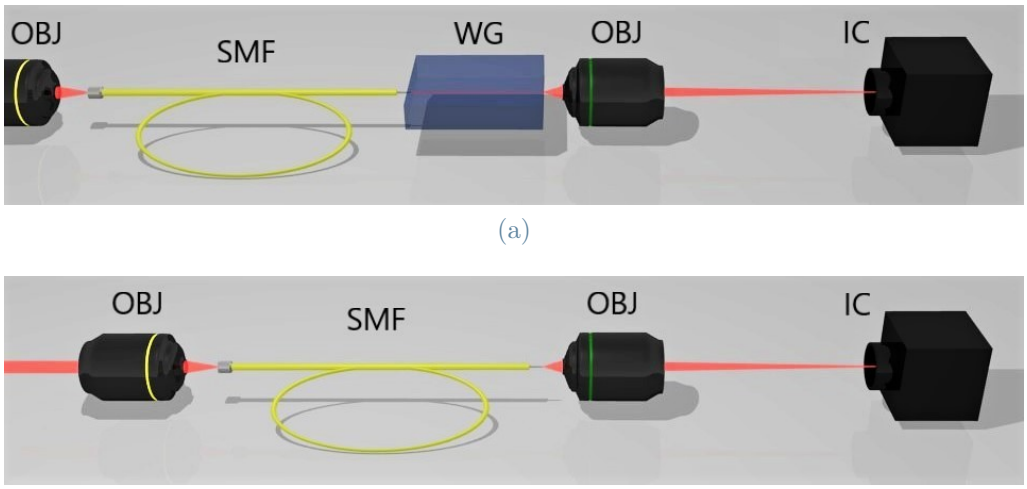


Figure 3.3: Setup for the acquisition of the mode images. a) for the acquisition of the waveguide's mode, the near-field image of the waveguide is focused on a CCD camera and acquired. b) To retrieve the fiber's mode, the sample is removed and the fiber tip is brought close to the output objective, taking care not to move the collection optics. The fiber mode has known dimensions which can be used for calibration.

to cover the whole dynamic range of the camera without saturating the response while maintaining a linear characteristic. Once the image is acquired, the sample is removed and the fiber and is brought close to the collection objective, which must remain in the same position as before in order to keep the magnification ratio constant. This way, the fiber mode intensity is acquired.
Finally, software analysis allows to compare the mode of the fiber with those of the waveguides, thus retrieving the waveguide mode dimensions once the fiber mode diameter is known.

Losses characterization

Light propagation inside a waveguide structure introduces losses. Their characterization is fundamental for photonics applications, especially when dealing with quantum applications which often require few-photon inputs.
First, the insertion losses (IL) of the device are characterized, defined as the total losses due to the presence of the circuit. IL are usually expressed in decibels:

$$IL_{dB} = 10 \log_{10} \left(\frac{P_{in}}{P_{out}} \right) \tag{3.1}$$

where P_{out} is the waveguide input power and P_{in} is the input power, i.e. the fiber output power. The characterization setup is the same described in the previous section, but the camera is substituted by a powermeter.

Insertion losses are the result of different pbysical processes leading to reflections or coupling with radiating modes, reducing the fundamental guided mode power:

$$IL_{dB} = 2FL_{dB} + CL_{dB} + PL_{dB/cm} \cdot l + BL_{dB/cm} \cdot l_c \tag{3.2}$$

the meaning of each term is explained in the following paragraphs.

Fresnel Losses Because of the index mismatch at the air-glass interface, the impinging light is partially back-reflected (Fresnel reflection). Fresnel losses (FL) are expressed in dB as

$$FL_{dB} = -10 \log_{10} \left[1 - \left(\frac{n_2 - n_1}{n_2 + n_1} \right)^2 \right] \tag{3.3}$$

where n_1 and n_2 are the refractive indeces of glass and air respectively. Since $n_1 \simeq 1.5$ in borosilicate glasses, FL$\simeq$ 0.18 dB per facet, so that they are accounted two times in 3.2.2.

Coupling Losses Coupling losses (CL) are due to the mismatch between the waveguide's fundamental mode $\mathbf{E}_{wg}(x, y)$ and the fiber's mode $\mathbf{E}_{in}(x, y)$, injected at the waveguide input. Within coupled mode theory, CL are defined with the superposition integral between the two modes - expressed in dB,

$$CL_{dB} = -10 \log_{10} \left(\frac{\left| \iint E_{wg} E_{in} \, dx \, dy \right|^2}{\iint \left| E_{wg} \right|^2 dx \, dy \cdot \iint \left| E_{wg} \right|^2 dx \, dy} \right) \tag{3.4}$$

Propagation Losses Light propagating in a material attenuates exponentially with propagation. Such propagation losses (PL) may be due to waveguide roughness, inhomogeneity in the core index profile, presence of defects or absorption centers can cause scattering and coupling with radiation modes.
Therefore, the optical power in a waveguide after a propagation distance l is $P(l) = P(0)e^{-\alpha l}$, where α is the attenuation coefficient and $P(0)$ is the initial power. PL are usually expressed in dB/cm

$$PL_{dB/cm} = -\frac{10}{l} \log_{10} \left(\frac{P(L)}{P(0)} \right) \tag{3.5}$$

so that the optical power is reduced by $PL_{dB/cm}$ after every cm of propagation. The relation between $PL_{dB/cm}$ and α is linear, namely $PL_{dB/cm} = 10\alpha_{cm^{-1}} \log_{10} e \simeq 4.3\alpha_{cm^{-1}}$.
Experimentally, PL are retrieved by measuring the output power of a straight waveguide for different lengths, cutting the device several times. However, this method is time consuming and damages the sample. Thus, one may simply invert equation 3.2.2 and obtain

$$PL_{dB/cm} = \frac{IL_{dB} - 2FL_{dB} - CL_{dB}}{l} \tag{3.6}$$

Although this method suffers from the statistical uncertainties related to the involved measured losses, it is more practical.

Bending Losses When a waveguide bends, the guided mode is distorted and partially coupled to a radiating mode. This is easy to picture in a ray optics description of the waveguide - indeed, when the waveguide bends the guided rays impinge on the core-cladding interface with a higher incidence angle, increasing their transmittance to the cladding. Such losses are found to decrease exponentially with the bending radius [112]:

$$BL_{dB/cm} = C_1 e^{-C_2 R} \tag{3.7}$$

where C_1 and C_2 are constants depending on the waveguide properties and mode profile (e.g. a higher index mismatch increases the mode confinement and reduces the bending losses).
Experimentally, BL are measured by comparing the insertion losses of a straight waveguide and a series of bent ones,each with different curvature radius but same bent path length l_c. Then, for every radius the bending losses are

$$\left(BL_{dB/cm}\right)_i = \frac{(IL_dB)_i - (IL_dB)_{straight}}{l_c} \tag{3.8}$$

This way, fitting the BL for different radii with the theoretical behavior one can extract the coefficients C_1 and C_2.

3.2.3. Characterization of the thermal shifters

Thermo-optic phase shifters exploit the temperature dependence of the refractive index to modulate the optical path in one of the two arms of the interferometer, imposing a phase shift between the two modes.
In particular, given coherent light is shone into one of the two input ports, the output optical power I_{out} measured at one of the two exits can be modulated by acting on the

phase difference φ between the two paths. Mathematically,

$$I_{out} = \frac{I_{max}}{2}\left[1 + \nu \cos\varphi\right]$$

where I_{max} is the sum of the output intensities from both ports and ν is the visibility of the interference fringe.
To act on the phase difference φ, Cr-Au resistive microheaters are exploited as described in [100].
Given the linearity of the heat equation and assuming a linear relation between refractive index and temperature change of the substrate, we can model the phase difference φ between the two arms as

$$\varphi = \Phi + \alpha P \tag{3.9}$$

where Φ is the static phase difference when the microheater is not biased, α is the tuning coefficient of the interferometer and P is the thermal power dissipated by the microheater, equivalent to the Joule effect power dissipated by the resistance. The electrical power is easily retrieved by electrical measurements with the known relation $P = VI$, where V and I are the voltage drop and current traversing the resistor, respectively.
Optimum performances are obtained when a large phase shift is imposed with low power dissipation. Then, maximizing the tuning coefficient α is necessary. An analytical expression for this parameter can be derived under suitable assumptions.
Considering an infinitesimal waveguide segment having length dl, the phase $d\varphi$ induced by the microheater is

$$d\varphi = \frac{2\pi}{\lambda}\Delta n(l)dl = \frac{2\pi n_t}{\lambda}\Delta T(l)dl \tag{3.10}$$

where $\Delta n(l)$ and $\Delta T(l)$ are the refractive index and tempreature difference between the two guides at the coordinate l, and we have introduced the thermo-optic coefficient of the substrate n_t. Here, we suppose the refractive index changes linearly with temperature. While this assumption is validated by experiments at room conditions, measurements at cryogenic temperatures show a more complex behaviour [113], for which a theoretical explanation has not yet been proposed.
If the linear relation holds, we can obtain α by integrating equation 3.2.3 over the entire optical path γ and substituting $\Delta\varphi = \varphi - \Phi$ into equation 3.2.3. Namely,

$$\alpha = \frac{\Delta\varphi}{P} = \frac{2\pi n_t}{\lambda}\int_\gamma \frac{\Delta T(l)}{P}dl \simeq \frac{2\pi n_t}{\lambda}\int_\gamma \frac{\Delta T(l)}{\bar{P}} = \frac{2\pi n_t}{\lambda}\int_\gamma \bar{R} \tag{3.11}$$

where we introduced the linear power dissipation density $\bar{P} = P/L$ and the thermal efficiency factor $\bar{R} = \Delta T/\bar{P}$.

While writing equation 3.2.3, we further assumed a the temperature difference $\Delta T(l)$ be constant over the entire extention of the microheater. This is valid whenever L is much greater than the transverse dimensions of the device - in this case, the thermal problem can be studied with a 2D geometry and the heat flux is always orthogonal to the light propagation, so that the temperature difference is constant and non-null only beneath the microheater.
The thermal efficiency factor $\bar{R}$ has the dimensions of a 2D resistance, and represents the only degree of freedom for the optimization of α. $\bar{R}$ can be increased by avoiding heat diffusion far from the waveguide, which includes using isolation microstructures or using the device in vacuum conditions.

Characterization of the thermo-optic phase shifters consists in evaluating the electrical power dissipated to impose a full 2π phase shift, which is directly related to α by

$$P_{2\pi} = \frac{2\pi}{\alpha} = \frac{\lambda}{n_t \bar{R}} \tag{3.12}$$

The performance of the micro-heaters is assessed with optical measurements both in room conditions and inside a vacuum chamber.

Characterization in air

A first characterization is carried out in room conditions to assess the working point of the resistors.
Light is coupled to the interferometer with the same setup described in Section 3.2.2, in the *end fire* configuration. Two 0.25-NA and 0.60-NA aspherical lenses are used for coupling the light onto the input facet and collecting the output light, respectively.

For the characterization of the *internal* phase shifters, a pair of powermeters (Ophir Nova II) is used to analyze the output intensities from both ends of the interferometer.
A DC voltage source (Keithley 2231A-30-3) is used to produce a tension sweep on the phase shifters. Since the transmission of either one of the two arms of the MZI is sinusoidal with the dissipated electrical power (see Section 4.5.4), we are able to determine the efficieny of the phase shifters.

For the characterization of the *external* phase shifters, the powermeters are substituted by a CCD camera to acquire a picture of the far-field image produced by the two outputs of the interferometer. The far-field pattern has a different spatial phase depending on the relative phase between the two arms (see Section 4.5.4). Analyzing the dependence of the phase on the dissipated electrical power, the efficiency of the phase shifter is evaluated.

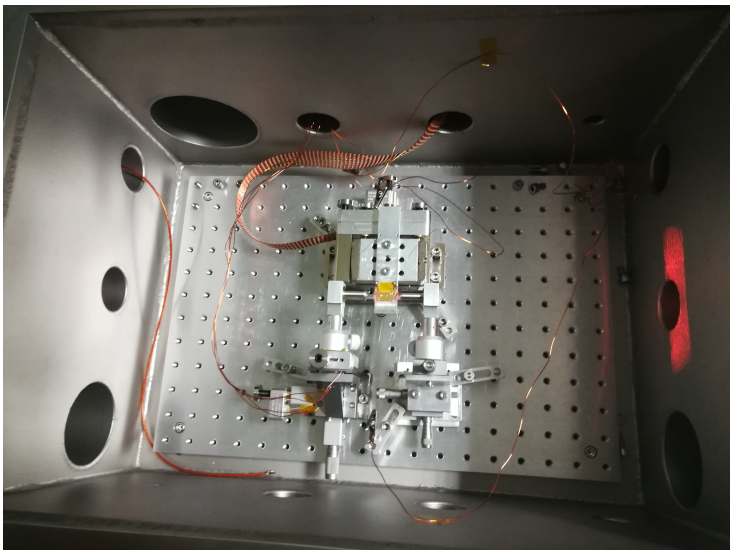


Figure 3.4: Setup for characterization in vacuum chamber. From left to right: the laser light enters the chamber from the middle pinhole, is then coupled by a 0.25-NA aspherical lens to the sample, then the output 0.40-NA aspherical lens collect the light and focuses it outside the vacuum chamber. The electrical nodes placed on the detectors are wired to the sourcemeter with copper connections.

Characterization in vacuum

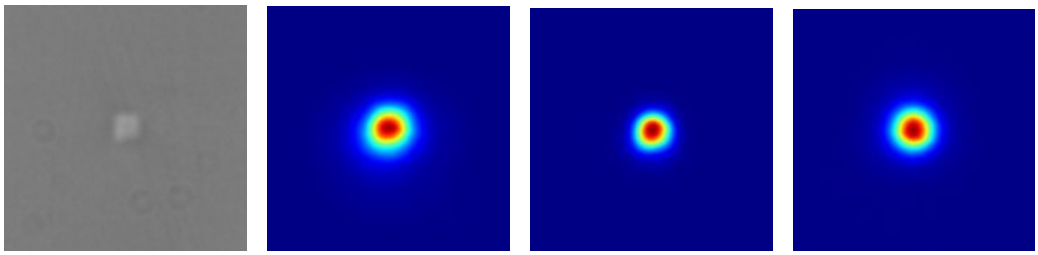
The characterization is repeated in vacuum conditions, since the global device will be kept in a vacuum chamber for the experiment.
Only a characterization of the *internal* phase shifters is necessary in order to asses the working point of the device. Indeed, all the phase shifters must be characterized again once the device is operational inside the cryostat due to the heavy dependence of the tuning coefficient on temperature in cryogenic conditions.

The setup is the same described in Section 3.2.3, only the sample is mounted inside a vacuum chamber as shown in Figure 3.4. Two 0.25-NA and 0.40-NA aspherical lenses are used for coupling the light onto the input facet and collecting the output light, respectively. Also, the DC voltage source is substituted by a sourcemeter (Keysight B2902A), which allows a more precise control of the tension in the low-power regime.

4 | Experimental Activity

4.1. Motivation and design of the experiment

Existing literature highlights how one of the advantages of waveguide quantum memories is the possibility to integrate the device with other photonic devices written by FLM [47, 50, 106, 107].
Within this context, this thesis work is aimed at realizing two fully reconfigurable Mach Zehnder interferometers (MZI) to be used together with a pair of rare earth-based quantum memories (QM) for the storage and retrieval of spatially entangled photons, for the realization of the Atomic Frequency Comb (AFC) protocol.
The QMs are constituted by a pair of Type I waveguides, inscribed by FLM inside a Pr^{3+}:Y_2SiO_5 crystal. Figure 4.1 shows the waveguide cross section and mode profile, together with the reference fiber mode.



(a) (b) (c) (d)

Figure 4.1: Type I waveguide inscribed in the crystal. a) Cross section of the waveguide. b,c) Horizontal (left) and vertical (right) modes of the waveguides. d) Reference mode of the fiber (S405-XP, Thorlabs).

The final device for the experiment is shown in figure 4.2.
Before the QM, a reconfigurable MZI allows to prepare the input photon in a superposition state of the kind

$$|\psi\rangle = \frac{|0\rangle + e^{j\phi}|1\rangle}{\sqrt{2}}$$

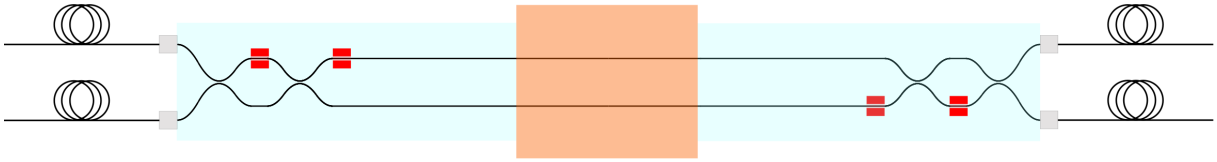


Figure 4.2: Schematics of the global interferometer. Two identical glass-written MZIs are glued to the Pr^{3+}:Y_2SiO_5 crystal containing the waveguide QMs. This resulting interferometer is then pigtailed to single-mode fibers on both ends. Thermo-optic phase shifters are implemented to grant reconfigurability of the circuit, togeter with bridge isolation structures (highlighted in red) to minimize heat dissipation in the sample.

where $|0\rangle$ and $|1\rangle$ represent the two paths of the photon, i.e. the two arms of the interferometer, and ϕ is the phase difference between the two paths.
In order to measure the output state of the QM, another reconfigurable MZI is placed after the crystal. The measurement process, commonly known as *quantum state tomography*, consists in assessing the output state coordinates on the Bloch sphere by executing simple projective measurements.

The necessity to exert tight control over the phase of the two paths poses some restraints on the MZI design.
First, the whole system must be integrated, i.e. both the MZI must be directly glued to the crystal waveguides. Only in this way can we avoid optical pathlength fluctuations that would hinder the phase control operations. This requires that the waveguide structures of the MZI shall have modes matched to those of the crystal.
Second, to realize all the projective measurements required for the quantum state tomography we must be able to fully extinct the transmission of one of the two modes of the output interferometer. This is only possible if the directional couplers that constitute the MZI are balanced, i.e. they grant a 50% splitting between bar and cross state.
Finally, since the device will be pigtailed to fibers on both ends we must grant good mode matching between the MZI mode and the fiber's as well.
Another constraint is given by the fact that the Pr^{3+}:Y_2SiO_5 crystal must be kept at cryogenic temperatures while operating, in order to preserve its coherence properties. This means that special care must be placed into minimizing heat dissipation inside the crystal. To do so, we implement bridge isolation microstructures (see Section 2.3.1) and place the heaters as far as possible from the crystal.

Reconfigurability of the interferometers allows to willingly modulate the bar and cross state transmission, and to apply an additional phase factor to the two modes.
To do so, we employ thermo-optic phase shifters, based on Joule effect heat dissipation

by a thin resistor deposited on top of the glass chip on opportune sections of the MZI. Namely, we make use of two phase shifter for every interferometer.
One phase shifter is placed in-between the two couplers of the MZI. This phase shifter - from here on called the *internal* shifter - allows to change the output transmission of the two modes of the MZI, creating the spatial entanglement state.
The other shifter is placed after the MZI. This *external* phase shifter allows to impose a controlled phase difference between the two modes. This is necessary both for the input chip to asses the validity of the protocol for any qubit, and for the output chip to perform the state tomography.

4.2. Optimization of the waveguide

In this Section, we present the experimental optimization process of waveguides and directional couplers, aimed at achieving low optical losses and high reproducibility.
Optimization of various parameters of the laser inscription needs to be performed, based on the particular application and substrate.
The integrated quantum memory stores photons with a wavelength of 607 nm - therefore, we need to design a single mode waveguide working at this wavelength. In particular, we need good coupling with both the optical fiber and the crystal waveguides, and low losses.

We focused on the optimization of three fabrication parameters: inscription power, translation speed and number of scans. Moreover, we tried fabricating with both water immersion objective and dry objective with adjustable depth of correction of spherical aberrations.. The reason for this is that water immersion reduces the index contrast between environment and glass sample (1.33/1.5 instead of 1/1.5 for dry fabrication), which means the focused beam will suffer less from aberration effects and the focusing will be more tight - this could prove beneficial for waveguides guiding modes at a wavelength as low as 607 nm. At the same time, dry objectives are simpler to use and preferable, when fabrication results are satisfactory. Other parameters are referred to previous experience of the research group with the same substrate.
Waveguides are fabricated with the setup described in section 3.1.1. The laser repetition rate is set to 1 MHz, with a pulse duration of 170 fs. Waveguides are inscribed in 1.1 mm thick alumino-borosilicate glass samples (EagleXG, Corning).
The waveguides are fabricated 30 µm below the sample surface in order to increase the efficiency of the thermo-optical shifters - placing the waveguide near the surface means less power is dissipated in the sample for a given refractive index modulation.

4.2.1. Losses optimization

For dry fabrications, we explored a wide range of irradiation powers from 290 mW to 470 mW, keeping the translation speed constant at 40 mm/s and trying 1, 3, 6, 15, 20 overlapping scans. The laser beam was focused using a 50× 0.45-NA microscope objective with compensation gear (Zeiss).
For water immersion fabrications, a smaller range of irradiation powers was explored from 180 mW to 300 mW. Also, different translation speeds were tested at 10, 15 and 20 mm/s, and 1, 3 and 9 overlapping scans. A 20× 0.50-NA water-immersion objective (Zeiss) was used.
The waveguides are characterized using a diode laser at 606 nm and the setup described in Section 3.2.2, in the *fiber-butt* configuration.

First, mode acquisition allowed to evaluate the overlap integral beween the mode of the optical fiber (S405-XP, ThorLabs) and that of waveguides, estimating the coupling losses. For dry fabrication, waveguides inscribed with powers higher than 390 mW showed a double guiding structure, unsuitable for our purposes, while single-mode behavior was obtained for lower powers.
For water immersion fabrication, the behaviour was less regular. While multiple guiding regions were observed for powers higher than 260 mW, multimode behavior was shown along the whole fabrication window, depending on the translation speed and number of scans.

Insertion losses are then measured, and the different loss terms - Fresnel losses, propagation Losses, coupling losses - are separated. Results for the best waveguides obtained are shown in Figure 4.3. Among the dry-fabricated waveguides, the best results were obtained with 390 mW writing power, 40 mm/s translation speed, 10 scans. Propagation losses were as low as 0.25 dB/cm, with coupling losses standing at 0.32 dB.
Among the WI-fabricated waveguides, a fabrication window around 195 mW was identified showing low coupling losses, while the best propagation losses were obtained around 240 mW. In both cases, high translation speed (20 mm/s) and moderate number of scans (3 or 6) proved best.
In the end, we chose the WI-fabricated waveguide at 240 mW, 20 mm/s and 3 scans, which showed the lowest value of propagation losses (0.13 dB/cm) while retaining low coupling losses (0.25 dB) [2].

[2]Low propagation losses are preferred over coupling losses, since we predict the final device to be long enough that the total propagation losses will surpass the constant coupling losses.

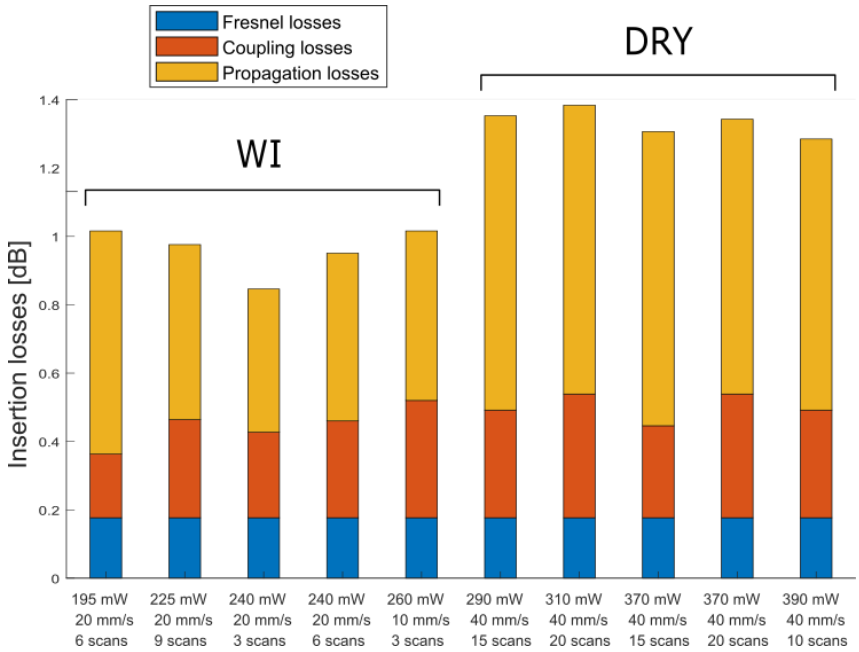
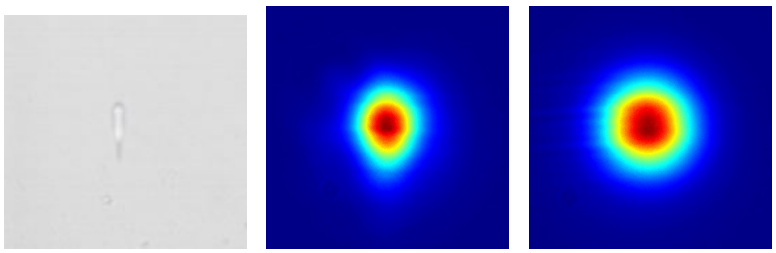


Figure 4.3: Insertion losses for the five best waveguides resulting from the optimization process both with dry fabrication and water immersion fabrication. The contribution of each loss term is highlighted, together with the fabrication parameters.



(a) (b) (c)

Figure 4.4: Optimized waveguide for single mode operation at 607 nm. a) Cross section of the best waveguide fabricated in the optimization process, after the annealing procedure. b) Waveguide mode, acquired with a CCD camera. c) Reference mode of the fiber (S405-XP, ThorLabs).

The post-annealing cross section of the waveguide, together with the image of its guided mode, are shown in Figure 4.4. No remarkable dependence of the waveguide mode profile was observed with respect to polarization of the guided light.

The final parameters of the design, adopted from now on through this work, are reported in Table 4.1.

Fabrication Parameters

Substrate	Eagle XG
Laser system	PHAROS
Repetition rate	1 MHz
Pulse duration	170 fs
Wavelength	1030 nm
Power	240 mW
Fabrication depth	30 µm
Writing speed	20 mm/s
Number of scans	3
Objective	20×, 0.50-NA WI

Table 4.1: Optimal fabrication parameters for single-mode waveguides at 607 nm.

4.3. Directional couplers

The next important step in the fabrication of our device is the optimization of the directional coupler modules. We hereby present a brief theoretical dissertation on coupled mode theory applied to directional coupler and discuss the experimental results in light of it.

4.3.1. Coupled-mode theory for directional couplers

When two waveguides are fabricated in close proximity, in such a way that there is a non-null overlap between their guided modes, a power exchange between them is possible, and this device is called Directional Coupler.
In the approximation of "weak coupling" regime - always valid in the case of laser written waveguides - the functioning of the directional coupler can be studied as follows. The modes of the coupled waveguide system can be written as a linear combination of the modes of the two individual waveguides, and the total field distribution is

$$E = A_1(z)E_1(x,y)e^{-i\beta_1 z} + A_2(z)E_2(x,y)e^{-i\beta_2 z} \tag{4.1}$$

where β_i is the propagation constant in the i-th waveguide, $A_i(z)$ is the longitudinal mode profile, $E_i(x,y)$ is the transverse mode profile, independent on the propagation direction.

Using coupled-mode theory, we can study the evolution of $A_1(z)$ and $A_2(z)$ [114]:

$$\begin{cases} \frac{dA_1}{dz} = -ik_{11}A_1 - ik_{12}A_2e^{i\Delta\beta z} \\ \frac{dA_2}{dz} = -ik_{21}A_1e^{-i\Delta\beta z} - ik_{22}A_2 \end{cases} \tag{4.2}$$

or in matrix form - defining $\mathbf{A}(z) = [A_1(z)\ A_2(z)]$

$$\frac{d\mathbf{A}}{dz} = \begin{bmatrix} -ik_{11} & -ik_{12}e^{i\Delta\beta z} \\ -ik_{21}e^{-i\Delta\beta z} & ik_{22} \end{bmatrix} \mathbf{A}(z) \tag{4.3}$$

where $\Delta\beta = \beta_1 - \beta_2$ and the k_{ij} are the coupling coefficients, defined by means of the overlap integral

$$k_{ij} = \frac{k_0}{2n_i} \frac{\iint E_i^* \Delta n_j(x,y)^2 E_j \ dx \ dy}{\iint E_i^* E_j \ dx \ dy} \tag{4.4}$$

where $\Delta n_j = \Delta n_j(x,y)$ is the difference between the index of the waveguide j and the total index profile in at *(x,y)*:

$$\Delta n_j(x,y) = n_j - n(x,y) \tag{4.5}$$

Assuming the index profile of the two single waveguides are identical, we have $\Delta\beta = 0$, which implies $k_{ij} = k_{ji} = k$ and $k_{ii} = k_{jj} = 0$. Equation 4.3.1 then simplifies to

$$\begin{cases} \frac{dA_1}{dz} = -ikA_2 \\ \frac{dA_2}{dz} = -ikA_1 \end{cases} \tag{4.6}$$

and the matrix equation 4.3.1 becomes

$$\frac{d\mathbf{A}}{dz} = -ik\mathbf{M}(z)\mathbf{A} \tag{4.7}$$

where

$$M(z) = \begin{bmatrix} 0 & k(z) \\ k(z) & 0 \end{bmatrix} \tag{4.8}$$

where we highlighted the dependence on z of the coupling coefficient. Indeed, coupling depends on the distance between the waveguides over the whole length of the structure. From this, the mode profile is easily retrieved once k is known by solving the first order differential equation, and

$$\mathbf{A}(z) = \mathbf{A}(z_0)e^{-i\int_{z_0}^{z} \mathbf{M}(s)ds} \tag{4.9}$$

and finally the power intensity distribution at a given z coordinate is given by $P_i(z) \propto |A_i(z)|^2$.
By opportunely choosing the length and the strength of the interaction, a directional coupler behaves as an optical power divider according to a desired splitting ratio.

In a real device, the waveguides are initially well separated well enough to consider them isolated and are then brought close to each other in the interaction region by curved waveguide paths. The phenomenon of light coupling start to take place before the waveguides reach their minimum interaction distance, and this fact has to be taken into account during the design of the device.
It can be shown that the coupling coefficient *k(z)* decays exponentially with the coupling distance *d*, since it depends on the overlap between the exponential tails of the fundamental modes of the isolated waveguides [115] :

$$k = ae^{-bd(z)} \tag{4.10}$$

where a and b are parameters and *d(z)* is the mutual distance between the waveguides, function of the propagation coordinate z.
Moreover, the overall reflectivity, defined as $R = P_{1,out}/(P_{1,out} + P_{2,out})$, depends on both the coupling length L and the coupling distance *d(z)*:

$$R = \cos^2(kL - \Phi_0) \tag{4.11}$$

where Φ_0 accounts for evanescent coupling.

In the following, we will use a notation common to integrated optics to talk about the behavior of the devices.
Assume we enter from the first input of the coupler, so that $P_{in} = P_{1,in}$. Then, the first output arm is called the *bar* state, so that $P_{1,out} = P_{bar}$, while the second output arm is the *cross* state, so that $P_{2,out} = P_{cross}$.
In this framework, the reflectivity R is equal to the normalized bar transmission of the coupler.

4.3.2. Coupling distance optimization

The circuit footprint is a key parameter for our device. Small devices show lower losses, plus compact devices are more suitable for cryostat operation.
To minimize the final device footprint, we decided to fabricate null-length couplers, where all the coupling is due to evanescent coupling in the bending regions. This way, the two

parameters to optimize are the bending radius R and the coupling distance $d(z)$.

Since the final device will only contain a total of four directional couplers, a full and complete characterization of the bending losses of our waveguides is not strictly necessary. In order to chose the smallest acceptable bending radius, we fabricated an array of directional couplers with different radii (15, 20, 30, 40, 50 and 60 mm) and different coupling distances (from 3 to 6 µm with a step of 0.5 µm).
Then, the total transmittance of each coupler was measured, defined as $\mathbf{T} = (P_{bar} + P_{cross})/P_{in}$. Results show how $\mathbf{T}$ grows when the R increases, whith no substantial improvement after R= 20 nm. This is in accordance with the theoretical behavior of bending losses, and with previous experience from the research group. Figure 4.5 shows the experimental data for the coupler's reflectivity at R= 20 mm, R= 30 mm, R= 40 mm and R= 50 mm for different coupling distances, together with a backward fit obtained with a simulation based on the coupled mode theory equations for the electric field distributions in the two waveguides. It was assumed the coupling coefficient $k(d)$ depends exponentially on the coupling distance d, as in 4.3.1.
We chose to proceed with bending radius R= 30 mm as a compromise between low curvature losses and circuit footprint.
From the backward fitting, we see how it's possible to obtain a balanced directional coupler by tuning the coupling distance in the range d = 4 µm - 4.5 µm. This fine tuning operation is performed on the same glass sample in which the final interferometers are fabricated, as will be discussed in the next section.

4.4. Mach-Zehnder interferometers

As already mentioned, the final device is a large interferometer composed of a first reconfigurable Mach-Zehnder interferometer (MZI), two waveguide quantum memories, and a final Mach-Zehnzer interferometer. Hence, we need to fabricate two identical devices in parallel.
From now on, we will refer to the input and output MZI as the one placed before the crystal and the one coming after, respectively. From a fabrication point of view, the two devices will remain indistinguishable until the electrical connections are soldered.

4.4.1. Design and characterization of the interferometers

To minimize the number of fabrication steps, we decided to optimize the coupling distance of the couplers and fabricate the corresponding MZI on the same glass sample. Furthermore, we decided to fabricate both the input and output chips starting from the

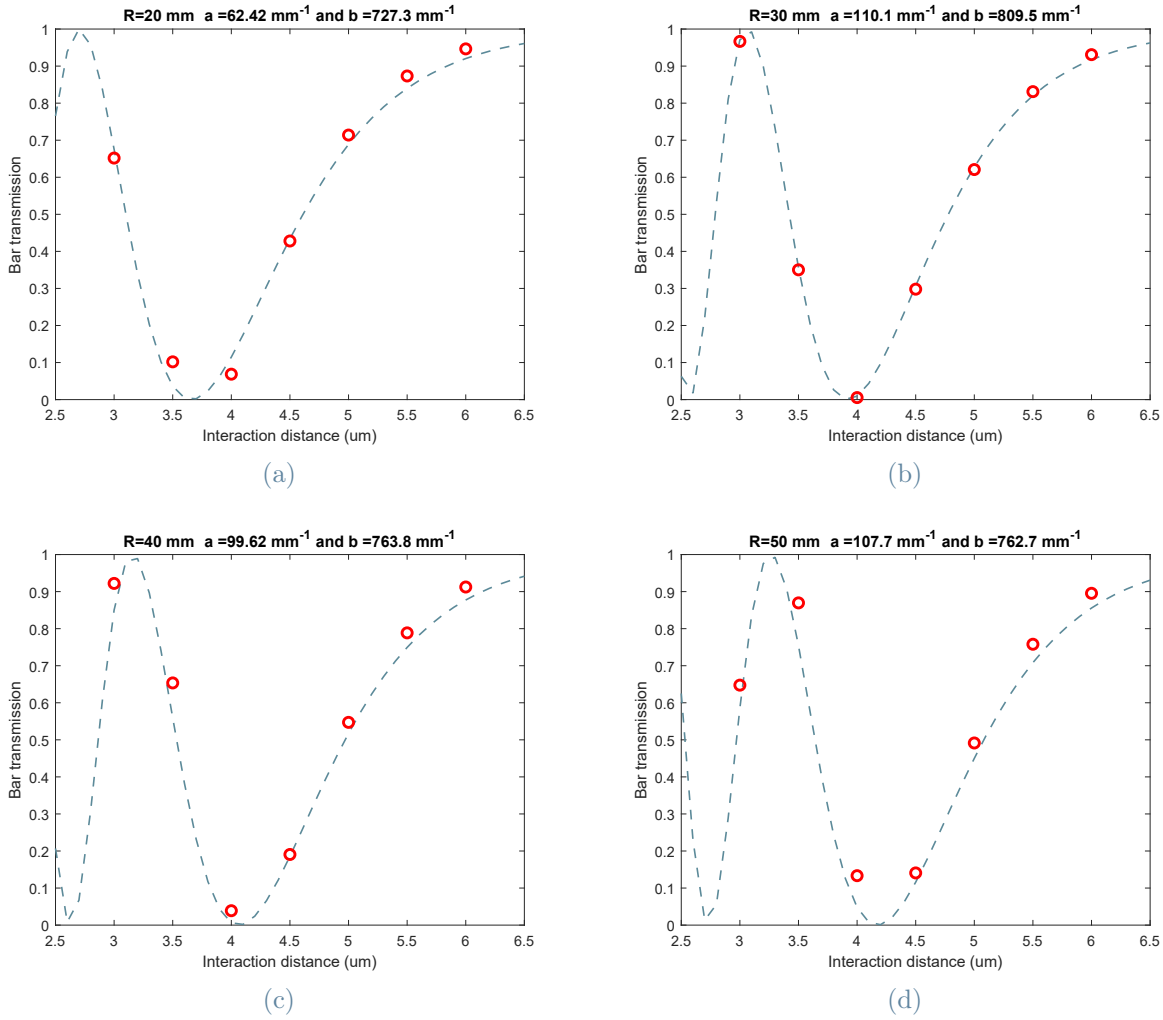
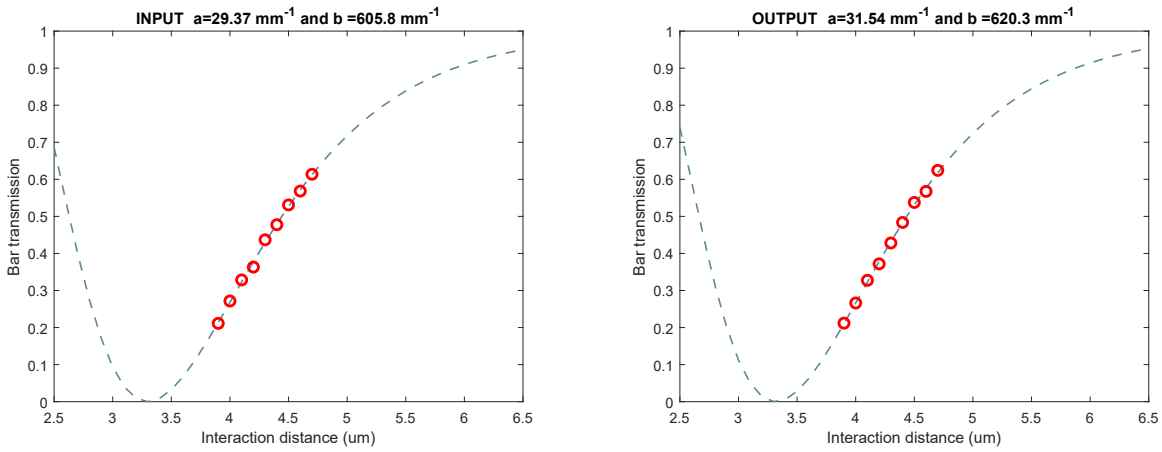


Figure 4.5: Experimental data of the reflectivity of the DC, as a function of the coupling distance and for curvature radius R=20 mm (a), R=30 mm (b), R=40 mm (c) and R=50 mm (d) for the input and output chip. The dashed line is a backward fit of the reflectivity, obtained according to the model described in Section 4.3.1.

same sample. Namely, we fabricated two identical circuits on the two halves of the glass sample. The sample was later separated in two parts.



(a) (b)

Figure 4.6: Fine tuning of the coupling distance d for couplers with curvature radius R= 30 mm. Experimental data together with the backward theoretical fit are shown for both the input (a) and output (b) chip.

For the optimization of the coupling distance we proceeded as in 4.3.2, fabricating 9 directional couplers with evenly spaced coupling distances between 3.9 µm and 4.7 µm. Figure 4.6 shows the experimental data and backward fit of the two sets of couplers. In both cases, we find that the coupler is balanced for a coupling distance d = 4.4 µm, with a bar transmission of approximately 48%.

Contextually, MZIs were fabricated for every coupling distance. Each interferometer has its modes separated by a pitch of 127 nm on the side dedicated to fiber pigtailing - in order to meet the pitch of commercial fiber arrays - while the side in contact with the crystal has a mode pitch of 80 nm. Each MZI is designed to accommodate two 1 mm long thermo-optic phase shifters, one between the two couplers (internal shifter) and one after the second shifter (external shifter), on the side near the crystal. The geometry of the final interferometer, together with the geometry of the isolation trenches, is summarized in Table 4.2.

The insertion losses of the two interferometers were measured, standing at 1.6 dB for the input MZI and 2 dB for the output MZI.

4.5. Thermo-optic phase shifters

After the fabrication of the interferometers, we proceeded with the design and realization of the thermo-optic phase shifters to control the internal and external phases of the devices. Indeed, a complete control of both phases becomes necessary for the operation of the final device, since there the external phases of the two single interferometers become internal phases of the global interferometer.
For this reason, two phase shifters were realized on each device, as shown in Figure 4.8.

4.5.1. Design of the microheaters

An accurate design of the microheaters is required to induce a full 2π phase shift with low power dissipation, which is crucial in our experiment since the device will be operating in a cryostat.
The electrical properties of the microheater are set by its geometry. Considering a rectangular heater, let h be its thickness, l its length and w the width. Then, the resistance is given by

$$R = \frac{\rho}{h}\frac{l}{w} \tag{4.12}$$

where ρ is the resistivity of the metal layer. Since the resistors are made of gold and the thickness is fixed at $h = 100$ nm by the deposition process, the quantity $\rho/h = 0.86\ \Omega/\square$ is fixed.
Different factors come into play for the choice of the value of l/w. First, we want the microheaters to have a small geometrical footprint, to keep the device as compact as possible. Moreover, we want to keep the microheaters as far from the crystal as possible, in order to avoid heat leakage in the quantum memory - this would heavily impair the realization of the quantum memory protocol, which requires cryogenic temperature to preserve coherence. On the other hand, a large resistor area would allow to dissipate the same amount of power with a lower current.
To solve the trade off, we decided to implement isolation trenches around the interferometer portion below the microheater.

4.5.2. Isolation trenches

The best possible results in terms of thermal isolation are achieved by using bridge trenches (see Section 2.3.1). Special care is required, since such structures are inherently more fragile than simple trench walls, since a long but thin stripe of glass containing the waveguide is suspended inside the ablated bridge.

From previous experience in the work group, we know that 1 mm long bridges with box thickness of 60 µm, separation of 20 µm between the boxes and a heigth of 60 µm provide good isolation together with sufficient mechanichal stability.

MZI geometry

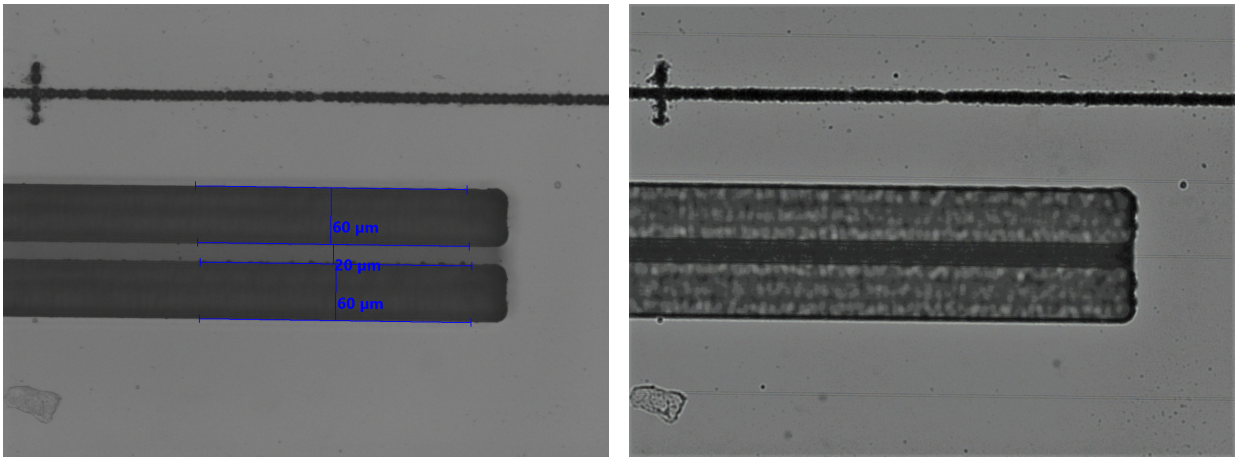
Fiber mode pitch	127 nm
Crystal mode pitch	80 nm
Coupling distance	4.4 µm
Trenches length	1 mm
Trenches width	60 µm
Trenches depth	60 µm
Trenches separation	20 mm
Device length	2.3 cm
Fabrication depth	30 µm

Table 4.2: Geometry of the final interferometers and isolation trenches.

Figure 4.7 shows microscope images of one of the two bridge ablations on one of the two chips. While the image acquired in transmission mode allows to check whether the waveguide is unimpaired by the bridge ablation, the one acquired in reflection mode clearly shows the ablation boundaries, allowing to determine the actual dimensions of the ablation.

4.5.3. Gold deposition and resistor patterning

After the isolation bridges are ablated, gold is deposited following the procedure described in Section 2.3.1.
Then, resistors are patterned using the same laser setup used for the other fabrication steps. In particular, ablation lines are inscribed on the gold surface in order to create thin isolated conductive paths - acting as resistances - right above the waveguide structures. The setup parameters for gold ablations are summarized in Table 4.3. The combined use of a low-aperture objective with the long wavelength allows to extend the focal volume of the laser beam by increasing its Rayleigh parameter, obtaining deeper ablations with better isolating power.



(a) (b)

Figure 4.7: Microscope images of one bridge ablation. a) Image acquired in reflection mode. The ablation boundaries are clearly defined, allowing for precise measurement of their dimensions. b) Image acquired in transmission mode. On the right side of the ablation we see the previously inscribed waveguide structure running parallel to the ablation, meaning that the bridge correctly encircles the waveguide.

Gold ablation parameters

Laser system	Yb-based cavity
Repetition rate	1 MHz
Pulse duration	170 fs
Wavelength	1030 nm
Power	200 mW
Writing speed	2 mm/s
Number of scans	1
Objective	10×, 15-NA dry

Table 4.3: Parameters of the gold ablation setup.

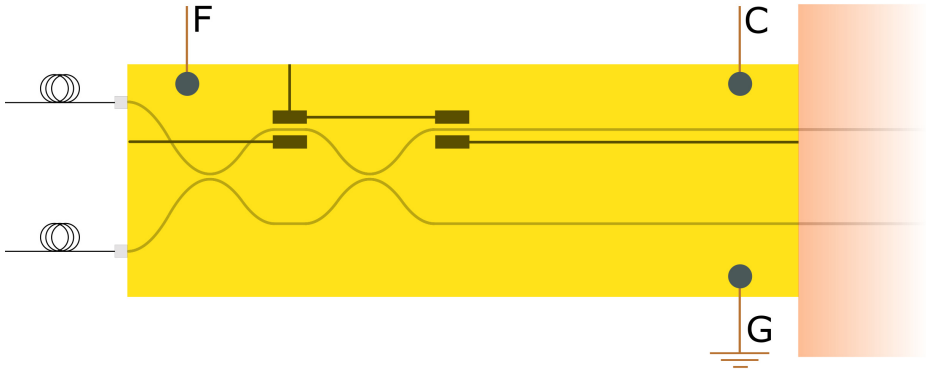
The ablations follow the scheme presented in Figure 4.8a. Copper wires are directly connected to the gold layer by tin soldering on the F, C and ground nodes. The **F** and **C** nodes are respectively close to the facets facing the fiber connections and the crystal, while the **G** node acts as common ground.
This way, we are able to act separately on the two resistances, constituted by the conduc-

tive paths **F-G** and **C-G**.
The value of the resistances can be estimated using equation 4.12: with l = 1mm and $w = 20$ µm, we obtain a nominal value of $R = 43$ Ω.
Testing with a multimeter shows how the actual values of the resistances differ from the ideal ones (see Table 4.4), possibly due to imperfections in the soldering or scratches on the gold surface.

Actual values of the ablated resistances

	R_{int}	R_{ext}	$R_{int} + R_{ext}$
Input chip	43 Ω	42.3 Ω	81.5 Ω
Output chip	44 Ω	44 Ω	86 Ω

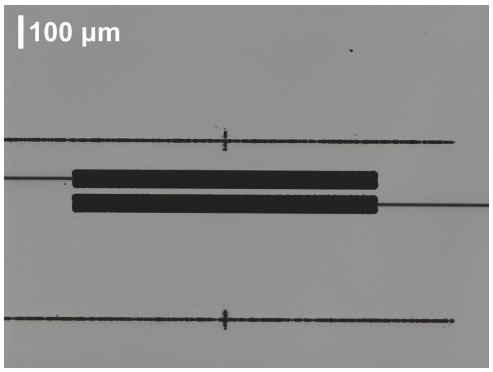
Table 4.4: Actual value of the electrical resistances in the two chips, measured with a multimeter.



(a)



(b)



(c)

Figure 4.8: Resistor patterning on the deposited gold film. a) Schematics of the electrical connections for the INPUT device. Three nodes (**F**,**C** and **G**) are isolated by ablating the gold surface, as highlighted by the dark brown lines. The thin gold layer between the isolation boxes acts as a resistance. b, c) Microscope images of the gold ablations, acquired in reflection mode.

A picture of the final chip is shown in Figure 4.9.

4.5.4. Characterization of the phase shifters

The next step is the characterization of the thermo-optic phase shifters. In particular, we are interested in the Phase-Power characteristic of each shifter, which ultimately tells how much electrical power is dissipated to produce a full 2π phase shift in the optical transmission profile.

For each chip, we need to analyze two phase shifters. The phase shifter located inside the MZI - between the two directional couplers - will from now on be called *internal* phase shifter, while the other one - outside the MZI, near the cristal facet - will be called *external*

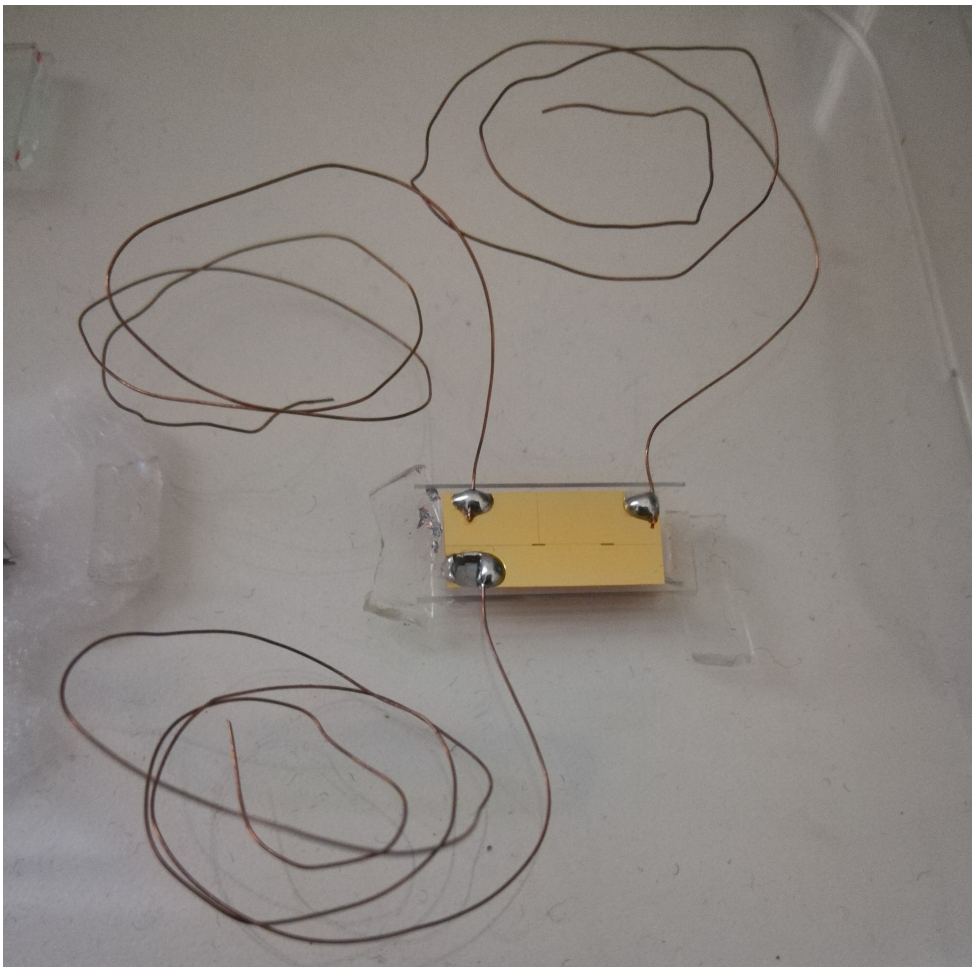


Figure 4.9: Photo of the output chip after all the fabrication steps. The isolation trenches and the ablation lines are visible in the darker portions on the gold layer. Copper wires are directly soldered on the metallic film.

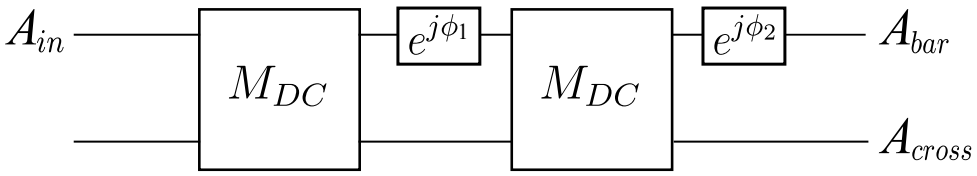


Figure 4.10: Matrix view of the interferometer with thermo-optic phase shifters.

phase shifter.

Internal phases For the single chip, only the internal phase shifter can produce a variation in the bar transmission. Indeed, a phase shift imposed before the second coupler alters the way the two modes interfere inside it, changing the transmission of the two modes.

This is easily understood when studying the MZI with the matrix formalism, starting from Figure 4.10. We assume that both couplers are balanced, so that their field transfer matrix is

$$M_{DC} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -j \\ -j & 1 \end{bmatrix} \tag{4.13}$$

Furthermore, we set the external phase $\phi_2 = 0$ and assume only one input P_{in}.

Then, the transfer matrix of the interferometer is

$$\begin{aligned} M_{MZI} &= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -j \\ -j & 1 \end{bmatrix} \begin{bmatrix} e^{j\phi_1} & 0 \\ 0 & 1 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -j \\ -j & 1 \end{bmatrix} = \\ &= \frac{1}{2} \begin{bmatrix} 1 & -j \\ -j & 1 \end{bmatrix} \begin{bmatrix} e^{j\phi_1} & -je^{j\phi_1} \\ -j & 1 \end{bmatrix} = \\ &= \frac{1}{2} \begin{bmatrix} e^{j\phi_1} - 1 & -j\left(e^{j\phi_1} + 1\right) \\ -j\left(e^{j\phi_1} + 1\right) & 1 - e^{j\phi_1} \end{bmatrix} = \\ &= -je^{j\phi_1/2} \begin{bmatrix} -\sin(\phi_1/2) & \cos(\phi_1/2) \\ \cos(\phi_1/2) & \sin(\phi_1/2) \end{bmatrix} \end{aligned} \tag{4.14}$$

so that the output amplitudes of the modes are

$$\begin{bmatrix} A_{bar} \\ A_{cross} \end{bmatrix} = -je^{j\phi_1/2} \begin{bmatrix} -\sin(\phi_1/2) & \cos(\phi_1/2) \\ \cos(\phi_1/2) & \sin(\phi_1/2) \end{bmatrix} \begin{bmatrix} A_{in} \\ 0 \end{bmatrix} \tag{4.15}$$

In the end

$$\begin{cases} A_{bar} = A_{in}\ je^{j\phi_1/2} \sin(\phi_1/2) \\ A_{cross} = A_{in}\ je^{j\phi_1/2} \cos(\phi_1/2) \end{cases} \tag{4.16}$$

If we now define the bar transmission T_{bar} as $T_{bar} = P_{bar}/(P_{bar} + P_{cross}) \propto |A_{bar}|^2$, we have

$$T_{bar} \propto \left|\sin\left(\frac{\phi_1}{2}\right)\right|^2 \tag{4.17}$$

and we see how the optical transmission is sinusoidal with the phase difference between the two modes.In particular, notice how there exists a certain condition for which the bar state is fully extincted, i.e. $T_{bar} = 0$.
If the two couplers were not balanced - as is our case - we would not be able to fully extinct the bar state. For this reason, a fully reconfigurable device can only be obtained by using balanced directional couplers. In particular, the fringe visibility of the bar transmission is a figure of merit for the reconfigurability of the device.

External phases On the other hand, a phase shift imposed after the interference took place only changes the relative phase between he two modes, not altering their intensity. Namely, if set ϕ_1 to have a balanced MZI, right before the external phase shifter we have $A_1 = A_2 = A/\sqrt{2}$. Then, after the external phase shifter:

$$\begin{bmatrix} A_{bar} \\ A_{cross} \end{bmatrix} = \begin{bmatrix} e^{j\phi_2} & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} A_1 \\ A_2 \end{bmatrix} \tag{4.18}$$

and

$$\begin{cases} A_{bar} = e^{j\phi_2} A/\sqrt{2} \\ A_{cross} = A/\sqrt{2} \end{cases} \tag{4.19}$$

which means that the power transmitted by the two modes is unaltered.
To study the effect of the external phase shifter, we will perform a far field analysis of the output light.
Namely, the Fourier transform of the source is produced on a plane far from it. Since in our case we have two point-like sources - the two exits of the MZI - their far field output will be a spatial sinusoid, with a phase dependent on the relative phase between the field profiles of the two modes.

The characterization setup is described in Section 3.2.3.
For each chip, we start by characterizing the behavior of the internal phase shifter. In particular, we plot the bar transmission as a function of the dissipated electrical power and calculate the electrical power needed to sweep a bar transmission period, following the theory exposed in Section 2.3.1.
First, we calculate a range of electrical powers to sweep. From previous experience, we know that our resistors can sustain up to ~ 30 mA DC before permanent damages to the

resistors themselves occour due to excessive heating. Following Joule's law, this clamps our maximum delivered power to ~ 35 mW, or a voltage of ~ 1.3 V. Since our sources are tension-controlled, we are interested in the voltage sweep.

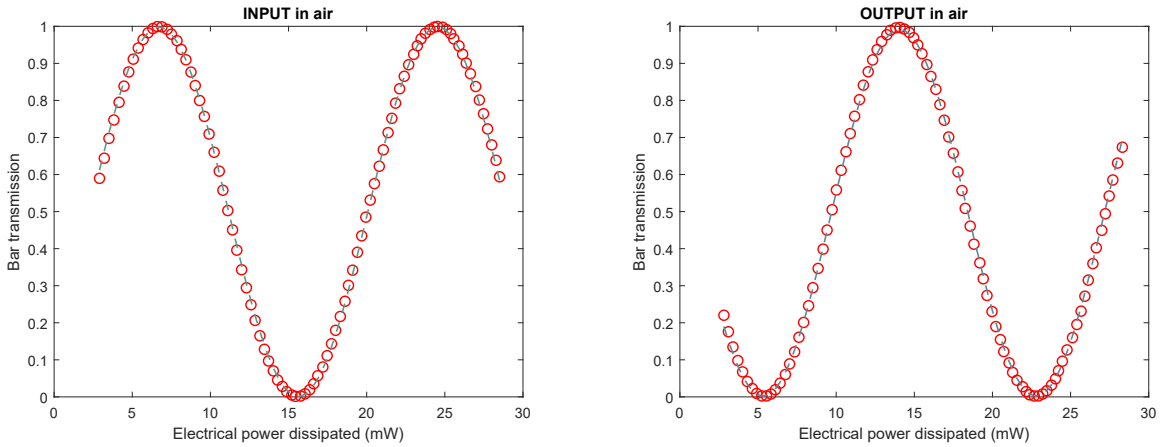
Figure 4.11 shows the normalized bar transmission profiles for the two chips, together with a sinusoidal fit that helps estimate the phase efficiency and bar extinction of each shifter.
Notice how the initial phase is different in the two chips. This is because the behaviour of the MZI is sensible to variations of the optical path in the coupling regions on the scale of the wavelength, being entirely determined by the interference effects between the two modes. In our case, this means that a variation of ~ 400 nm [3] on the optical path, e.g. due to temperature fluctuations during the fabrication, would cause a full 2π shift in the interference pattern of the coupler.
Numerical results are summarized in Table 4.5. We report bar extinction of few permille points, with phase efficiencies of 17.8 mW/2π and 17.5 mW/2π for the two chips.
Discrepancies in the phase efficiencies can be traced back to the fact that the resistor values for each chip were different to begin with (see Table 4.4). Indeed, resistance changes with temperature following a linear behaviour

$$R(T) = R_0 + \alpha T$$

so that heating is different in the two cases.

Next, the external shifters are studied. For this, we impose a tension on the internal shifter such that the output intensities of the two modes are balanced, and sweep the tension on the external shifter to carry out the far field analysis.
Namely, we set the tension on the external shifter to impose a phase difference on one of the two modes, and study how the interference pattern produced by the two modes changes by varying the applied tension. In particular, we observe how the spatial phase of the interference pattern changes, and we are able to produce a linear fit to determine the phase efficiency.

[3] The wavelength of 607 nm must be divided by the refractive index of borosilicate glass to calculate the optical path. Since $n_{glass} \simeq 1.5$, a full period shift happens with a variation of 607 nm/1.5$\simeq$ 400 nm.



(a) (b)

Figure 4.11: Bar transmission of the input (a) and output (b) chip as a function of the electrical dissipated power, measured in room conditions. The sine fit (dashed blue line) allows to extract the efficiency of the interal phase shifter of each interferometer, in the form of its tuning coefficient.

Phase shifters characterization in air

	Internal efficiency	External efficiency	Bar extinction
Input chip	17.8 mW/2π	18.0 mW/2π	1 μW/mW
Output chip	17.5 mW/2π	18.5 mW/2π	2 μW/mW

Table 4.5: Values of the phase efficiencies for the internal and external phase shifters of both chips, measured in room conditions.

Experimental results together with the linear fit are shown in Figure 4.12. We report phase efficiencies of 18 mW/2π for the input device and 18.6 mW/2π for the output device.

Finally, the internal phases are characterized in vacuum. The setup as described in Section 3.2.3 is shown in Figure 3.4. Since thermal isolation is more effective in vacuum than in air, we sweep the sourcemeter tension over a more tight range in order to avoid heating the resistors to the point of breaking them.
At the time this thesis is being written, only the Input chip has been characterized in the vacuum chamber.
Results are shown in Figure 4.13. We report a phase efficiency of $\sim$ 1.5 mW/2π, one order of magnitude lower than the one measured in air. Fringe visibility is again 1 μW/mW, in

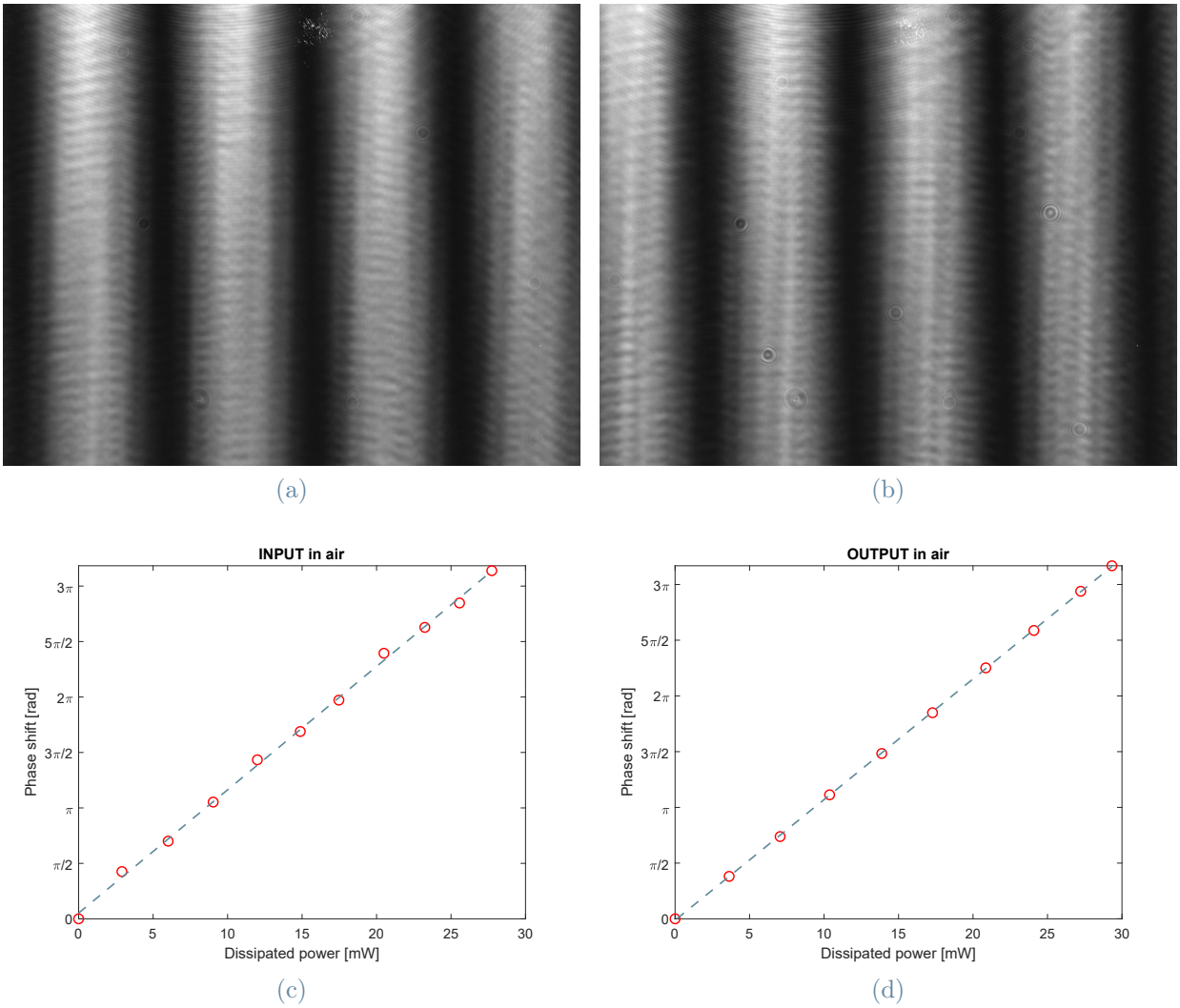


Figure 4.12: Characterization of the external phase shifters. a), b) Far field images of the MZI outputs for different electrical power dissipated on the external phase shifter. External phases of the input (c) and output (d) devices measured in room conditions. The phases have been calculated by extrapolating the phase shift of the far-field pattern produced by the MZI output, considering null-phase for no power dissipated. A linear fit (dashed blue line) allows to retrieve the phase efficiency as the slope of the line.

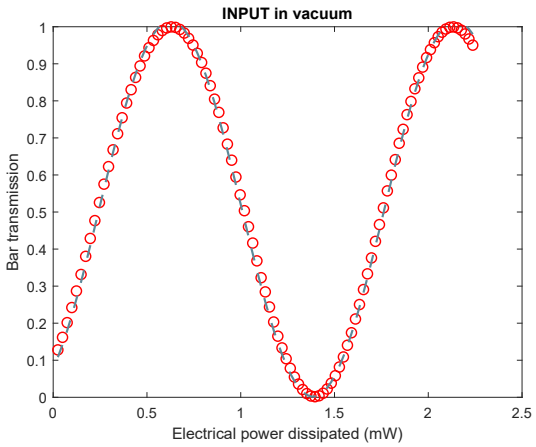


Figure 4.13: Experimental data for the bar transmission of the input MZI in vacuum conditions (in red), together with a sinusoidal fit (dashed blue line) to extract the phase efficiency.

accordance with the previous measurements - being only a property of the MZI, we did not expect it to change with the environment conditions.

5 | Conclusions and Future Perspectives

In this thesis work we have presented the fabrication by femtosecond laser writing of two fully-reconfigurable interferometers to work at 607 nm, together with their characterization. The optimization of the fabrication parameters was first addressed, highlighting how water immersion allows to obtain lower losses with respect to standard dry fabrication at the operating wavelength. We also found how the multi-scan approach allows to obtain smoother modifications. Moreover, an annealing process was implemented to further reduce optical losses. The optical losses were experimentally extimated to be lower than 0.2 dB/cm for single-mode guidance at 607 nm. This helped us fabricate high quality directional couplers, obtaining a 48/62 splitting coupler with curvature radius 30 mm, coupling distance 4.3 µm and null coupling length. Starting from this, we realized a pair of identical chips containing a Mach Zehnder interferometer made of two directional couplers and characterized their losses. Afterwards, we implemented a pair of thermo optic phase shifters on each chip. First, isolation bridge microstructures were fabricated by laser ablation in each interferometer, encasing two sections of of one arm. Then, a gold layer was deposited on top of the device, and resistors were patterned by laser ablation. Electrical connections were soldered on the gold film, and the phase shifters were characterized both in room conditions and inside a vacuum chamber.

The chip pair will be glued on the two ends of a pair of laser-written waveguide quantum memories inside a Pr^{3+}:Y_2SiO_5 crystal. In turn, the whole device will be pigtailed to single-mode optical fibers to obtain a fiber-integrated, fully reconfigutable optical quantum memory. The device will be used in the group of prof Hughues de Riedmatten at ICFO (Barcelona) for performing storage of photonic entanglement following the AFC protocol.

Future work will follow two parallel paths. Improvement of the waveguide fabrication in the crystal substrate is necessary to extend the AFC scheme to the spin wave and realize an on-demand quantum memory. On the other hand, larger reconfigurable devices will

be studied for the integration with quantum memories, to explore other applications e.g. linear optical quantum computing or complex quantum repeater links.

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